Structural fault dictionaries of the series–parallel networks of Ladenheim's catalogue

99 networks, 798 single hard faults, one page per network

Juraj Benić[1], Mario Essert[2], Ivana Kuzmanović Ivičić[1], Marko Orešković[3]

[1]School of Applied Mathematics and Informatics, University of Osijek, Osijek, Croatia
[2]Retired, formerly Faculty of Mechanical Engineering and Naval Architecture, University of Zagreb, Zagreb, Croatia
[3]National and University Library, Zagreb, Croatia

November 3, 2026

How the dictionaries were obtained. Every network is read as an *m-number* in the sense of Šare's m-theory: the elements C, R, L are the letters A, B, C of a three-letter alphabet and the series and parallel connections are the two additions of the m-system. From the m-word of a network its signature, the m-quadruple $\Phi = (v(l), \lambda_s, \lambda_p, v(r))$, is read off; it is the exponent structure of the impedance (orders at $s = 0$ and $s = \infty$, degrees of numerator and denominator). Each single hard fault is obtained by replacing one element by an ideal short or open in the m-expression and simplifying, so the whole dictionary – detectability of every fault and the ambiguity groups – follows from the schematic alone, with no element values and no simulation. All m-word computations were done with the Python package `mnumbers`[1] (version 0.1.0); the dictionaries and this document were generated by the scripts `mexpr_re.py` and `make_report.py`.

Detectability of the single hard faults

n	networks	faults	port	DC	HF	sweep	str. undet.
1	3	6	6	0	0	0	0
2	6	24	12	6	6	0	0
3	16	96	20	22	20	32	2
4	34	272	36	52	48	120	16
5	40	400	32	48	48	236	36
all	99	798	106	128	122	388	54
			13.3%	16.0%	15.3%	48.6%	6.8%

Ambiguity groups

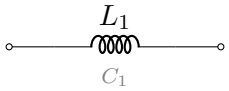
			structural			absolute		str. only	signature
n	networks	faults	groups	faults	largest	groups	faults	faults	faults
1	3	6	0	0	–	0	0	0	0
2	6	24	6	12	2	6	12	0	12
3	16	96	16	42	3	16	34	8	42
4	34	272	52	148	4	48	98	50	160
5	40	400	72	208	6	64	128	80	282
all	99	798	146	410	6	134	272	138	496
				51.4%			34.1%	17.3%	62.2%

[1]`https://github.com/JurajBenic/mNumbers`

Notation

- Elements: resistor R (m-letter B), inductor L (m-letter C), capacitor C (m-letter A). The grey labels under the elements in the schematics are the labels used in the tables.
- $x \,\mathsf{s}\, y$ is the series and $x \,\mathsf{p}\, y$ the parallel connection; K is a short circuit ($Z = 0$) and P an open circuit ($Z = \infty$).
- m-word: the raw word of the network as computed by the s- and p-addition of the letters, followed by its canonical form (the unique irreducible reduct written in capital letters). Words equal to it as m-numbers by the reversal rule are listed in parentheses.
- Signature $\Phi = (v(l), \lambda_s, \lambda_p, v(r))$; the impedance is $Z(s) = s^{v(l)}\, P(s)/Q(s)$ with $\deg P = \lambda_p$, $\deg Q = \lambda_s$, $\lambda = \lambda_s + \lambda_p$ the number of corner frequencies.
- Corners: $\delta_0 = Z(0)$ ($A \to$ P, $C \to$ K), $\delta_\infty = Z(\infty)$ ($A \to$ K, $C \to$ P), δ_{K} ($A, C \to$ K), δ_{P} ($A, C \to$ P).
- Faults: every element shorted (K, "short") and opened (P, "open"). Detected by: *DC* if $v(l)$ changes, *HF* if $v(r)$ changes, *sweep* if λ changes, *port* if the port itself is shorted or opened. A fault is *structurally undetectable* (column "str. undet.") when its signature equals the nominal one, so that none of the three signature tests (DC, HF, corner count) detects it for any choice of element values; for every such fault in the catalogue the m-word of the faulty network also equals the nominal one as an m-number. Such a fault may still be detectable through value-dependent features of the response, such as a change of level or of damping.
- Ambiguity groups. *Structural*: the residual networks have the same topology and element types after merging of same-type neighbours; no value-free test separates the members. *Absolute*: the sub-groups of a structural group consisting of all opens of one series node or all shorts of one parallel node; their residual networks are identical, element values included, and no port measurement can tell the faults apart. A structural group may coincide with an absolute group, contain absolute groups together with merged faults (members outside an absolute group differ in element values), or contain no absolute pair; "str. only" counts the faults of structural groups that are in no absolute group. *Signature*: the faults share the quadruple Φ (or collapse the port in the same way); *nominal* denotes the fault-free network.
- In the summary tables "DC" counts the faults that change $v(l)$ (in a few cases also $v(r)$), "HF" those that change $v(r)$ only, "sweep" those that change only λ, and "str. undet." the structurally undetectable faults.

Network 1



Network

realisation	`L`
m-expression	C_1
m-word	`C` $\rightarrow$ `C`
signature	$\Phi = (1, 0, 0, 1), \quad Z(s) = s^1\, P/Q, \quad \deg P = 0, \quad \deg Q = 0, \quad \lambda = 0$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

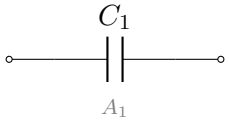
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	K	`K`	–	port	
C_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	none
absolute (identical residual networks)	none
signature (same quadruple)	none

Network 2



Network

realisation	`C`
m-expression	A_1
m-word	`A` $\rightarrow$ `A`
signature	$\Phi = (-1, 0, 0, -1), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 0, \quad \deg Q = 0, \quad \lambda = 0$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

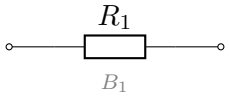
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	K	`K`	–	port	
A_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	none
absolute (identical residual networks)	none
signature (same quadruple)	none

Network 3



Network

realisation	`R`
m-expression	B_1
m-word	`B` $\rightarrow$ `B`
signature	$\Phi = (0,0,0,0), \quad Z(s) = s^0\,P/Q, \quad \deg P = 0, \quad \deg Q = 0, \quad \lambda = 0$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1, \quad \delta_{\mathrm{K}} = B_1, \quad \delta_{\mathrm{P}} = B_1$

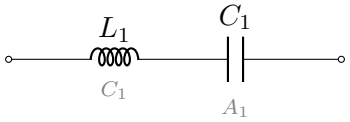
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	none
absolute (identical residual networks)	none
signature (same quadruple)	none

Network 4



Network

realisation	`L s C`
m-expression	$C_1 \,\mathtt{s}\, A_1$
m-word	`ACcAaC` $\rightarrow$ `AC`
signature	$\Phi = (-1, 0, 2, 1), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 0, \quad \lambda = 2$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

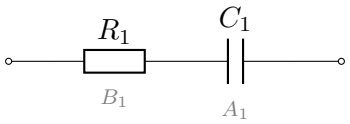
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	A_1	`A`	$(-1, 0, 0, -1)$	HF, sweep	
C_1	open	P	`P`	–	port	
A_1	short	C_1	`C`	$(1, 0, 0, 1)$	DC, sweep	
A_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	$\{C_1$ open, A_1 open$\}$
absolute (identical residual networks)	$\{C_1$ open, A_1 open$\}$
signature (same quadruple)	$\{C_1$ open, A_1 open$\}$

Network 5



Network

realisation	`R s C`
m-expression	$B_1 \,\mathtt{s}\, A_1$
m-word	`ABbAaB` $\rightarrow$ `AB`
signature	$\Phi = (-1, 0, 1, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 1, \quad \deg Q = 0, \quad \lambda = 1$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

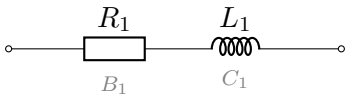
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	A_1	`A`	$(-1, 0, 0, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	DC, sweep	
A_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, A_1 open$\}$
signature (same quadruple)	$\{B_1$ open, A_1 open$\}$

Network 6



Network

realisation	`R s L`
m-expression	$B_1\,\mathtt{s}\,C_1$
m-word	`BBcbCC` $\rightarrow$ `BC`
signature	$\Phi = (0,0,1,1), \quad Z(s) = s^0\,P/Q, \quad \deg P = 1, \quad \deg Q = 0, \quad \lambda = 1$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

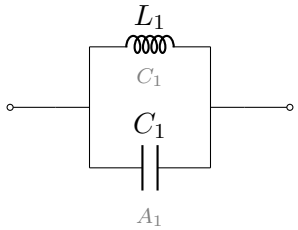
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	C_1	`C`	$(1,0,0,1)$	DC, sweep	
B_1	open	P	`P`	–	port	
C_1	short	B_1	`B`	$(0,0,0,0)$	HF, sweep	
C_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, C_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open$\}$
signature (same quadruple)	$\{B_1$ open, C_1 open$\}$

Network 7



Network

realisation	`L p C`
m-expression	C_1 `p` A_1
m-word	`CCacAA` $\rightarrow$ `CA`
signature	$\Phi = (1, 2, 0, -1)$, $Z(s) = s^1\, P/Q$, $\deg P = 0$, $\deg Q = 2$, $\lambda = 2$
corners	$\delta_0 = \mathrm{K}$, $\delta_\infty = \mathrm{K}$, $\delta_\mathrm{K} = \mathrm{K}$, $\delta_\mathrm{P} = \mathrm{P}$

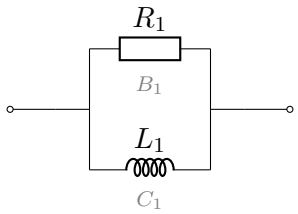
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	K	`K`	–	port	
C_1	open	A_1	`A`	$(-1, 0, 0, -1)$	DC, sweep	
A_1	short	K	`K`	–	port	
A_1	open	C_1	`C`	$(1, 0, 0, 1)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{C_1 short, A_1 short}
absolute (identical residual networks)	{C_1 short, A_1 short}
signature (same quadruple)	{C_1 short, A_1 short}

Network 8



Network

realisation	`R p L`
m-expression	$B_1 \,\mathtt{p}\, C_1$
m-word	`CBbCcB` $\rightarrow$ `CB`
signature	$\Phi = (1,1,0,0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 0, \quad \deg Q = 1, \quad \lambda = 1$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

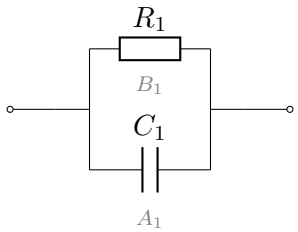
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	C_1	`C`	$(1,0,0,1)$	HF, sweep	
C_1	short	K	`K`	–	port	
C_1	open	B_1	`B`	$(0,0,0,0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, C_1 short$\}$
absolute (identical residual networks)	$\{B_1$ short, C_1 short$\}$
signature (same quadruple)	$\{B_1$ short, C_1 short$\}$

Network 9



Network

realisation	`R p C`
m-expression	B_1 `p` A_1
m-word	`BBabAA` $\rightarrow$ `BA`
signature	$\Phi = (0, 1, 0, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 0, \quad \deg Q = 1, \quad \lambda = 1$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

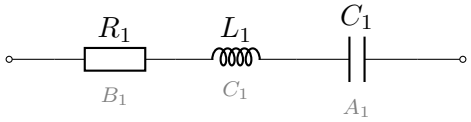
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	A_1	`A`	$(-1, 0, 0, -1)$	DC, sweep	
A_1	short	K	`K`	–	port	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 short, A_1 short}
absolute (identical residual networks)	{B_1 short, A_1 short}
signature (same quadruple)	{B_1 short, A_1 short}

Network 10



Network

realisation	`R s L s C`
m-expression	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, A_1$
m-word	`ABBcbCCcAaC` $\rightarrow$ `AC`
signature	$\Phi = (-1, 0, 2, 1), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 0, \quad \lambda = 2$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

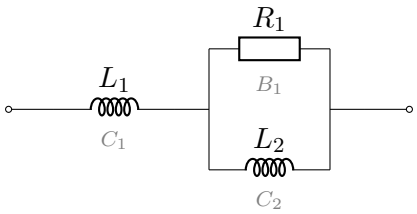
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathtt{s}\, A_1$	`AC`	$(-1, 0, 2, 1)$	none	✓
B_1	open	P	`P`	–	port	
C_1	short	$B_1 \,\mathtt{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	HF, sweep	
C_1	open	P	`P`	–	port	
A_1	short	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	DC, sweep	
A_1	open	P	`P`	–	port	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, C_1 open, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open, A_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_1 short$\}$ $\{B_1$ open, C_1 open, A_1 open$\}$

Network 11



Network

realisation	`L s (R p L)`
m-expression	$C_1 \,\mathtt{s}\, (B_1 \,\mathtt{p}\, C_2)$
m-word	`CCCCBbCcBC` $\rightarrow$ `CBC`
signature	$\Phi = (1,1,1,1), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

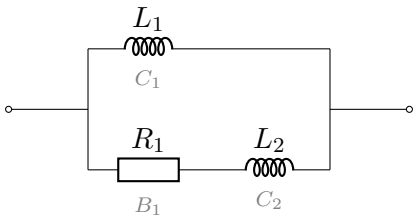
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	$B_1 \,\mathtt{p}\, C_2$	`CB`	$(1,1,0,0)$	HF, sweep	
C_1	open	P	`P`	–	port	
B_1	short	C_1	`C`	$(1,0,0,1)$	sweep	
B_1	open	$C_1 \,\mathtt{s}\, C_2$	`C`	$(1,0,0,1)$	sweep	
C_2	short	C_1	`C`	$(1,0,0,1)$	sweep	
C_2	open	$C_1 \,\mathtt{s}\, B_1$	`BC`	$(0,0,1,1)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 short, B_1 open, C_2 short}
absolute (identical residual networks)	{B_1 short, C_2 short}
signature (same quadruple)	{B_1 short, B_1 open, C_2 short}

Network 12



Network

realisation	`L p (R s L)`
m-expression	$C_1 \,\mathtt{p}\, (B_1 \,\mathtt{s}\, C_2)$
m-word	`CCbcBBcbCCC` $\rightarrow$ `CBC`
signature	$\Phi = (1,1,1,1), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

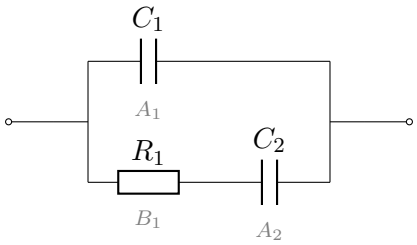
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	K	`K`	–	port	
C_1	open	$B_1 \,\mathtt{s}\, C_2$	`BC`	$(0,0,1,1)$	DC, sweep	
B_1	short	$C_1 \,\mathtt{p}\, C_2$	`C`	$(1,0,0,1)$	sweep	
B_1	open	C_1	`C`	$(1,0,0,1)$	sweep	
C_2	short	$C_1 \,\mathtt{p}\, B_1$	`CB`	$(1,1,0,0)$	HF, sweep	
C_2	open	C_1	`C`	$(1,0,0,1)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 short, B_1 open, C_2 open}
absolute (identical residual networks)	{B_1 open, C_2 open}
signature (same quadruple)	{B_1 short, B_1 open, C_2 open}

Network 13



Network

realisation	`C p (R s C)`
m-expression	$A_1\ \mathtt{p}\ (B_1\ \mathtt{s}\ A_2)$
m-word	`AAAABbAaBA` $\rightarrow$ `ABA`
signature	$\Phi = (-1, 1, 1, -1), \quad Z(s) = s^{-1}\,P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

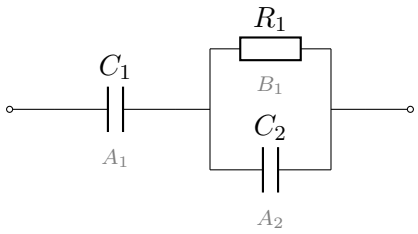
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	K	`K`	–	port	
A_1	open	$B_1\ \mathtt{s}\ A_2$	`AB`	$(-1, 0, 1, 0)$	HF, sweep	
B_1	short	$A_1\ \mathtt{p}\ A_2$	`A`	$(-1, 0, 0, -1)$	sweep	
B_1	open	A_1	`A`	$(-1, 0, 0, -1)$	sweep	
A_2	short	$A_1\ \mathtt{p}\ B_1$	`BA`	$(0, 1, 0, -1)$	DC, sweep	
A_2	open	A_1	`A`	$(-1, 0, 0, -1)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, B_1 open, A_2 open$\}$
absolute (identical residual networks)	$\{B_1$ open, A_2 open$\}$
signature (same quadruple)	$\{B_1$ short, B_1 open, A_2 open$\}$

Network 14



Network

realisation	`C s (R p C)`
m-expression	A_1 `s` $(B_1$ `p` $A_2)$
m-word	`AAbaBBabAAA` $\rightarrow$ `ABA`
signature	$\Phi = (-1, 1, 1, -1)$, $Z(s) = s^{-1}\,P/Q$, $\deg P = 1$, $\deg Q = 1$, $\lambda = 2$
corners	$\delta_0 = \mathrm{P}$, $\delta_\infty = \mathrm{K}$, $\delta_\mathrm{K} = \mathrm{K}$, $\delta_\mathrm{P} = \mathrm{P}$

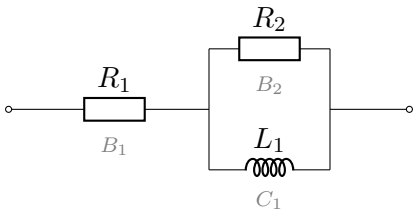
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	B_1 `p` A_2	`BA`	$(0, 1, 0, -1)$	DC, sweep	
A_1	open	P	`P`	–	port	
B_1	short	A_1	`A`	$(-1, 0, 0, -1)$	sweep	
B_1	open	A_1 `s` A_2	`A`	$(-1, 0, 0, -1)$	sweep	
A_2	short	A_1	`A`	$(-1, 0, 0, -1)$	sweep	
A_2	open	A_1 `s` B_1	`AB`	$(-1, 0, 1, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 short, B_1 open, A_2 short}
absolute (identical residual networks)	{B_1 short, A_2 short}
signature (same quadruple)	{B_1 short, B_1 open, A_2 short}

Network 15



Network

realisation	`R s (R p L)`
m-expression	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_1)$
m-word	`BBcbCBbCcBB` $\rightarrow$ `BCB`
signature	$\Phi = (0,1,1,0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathtt{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathtt{s}\, B_2$

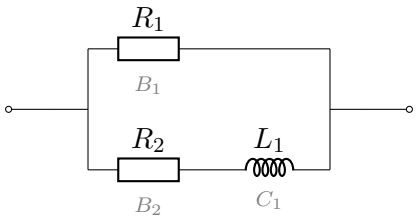
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathtt{p}\, C_1$	`CB`	$(1,1,0,0)$	DC, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0,0,0,0)$	sweep	
B_2	open	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0,0,1,1)$	HF, sweep	
C_1	short	B_1	`B`	$(0,0,0,0)$	sweep	
C_1	open	$B_1 \,\mathtt{s}\, B_2$	`B`	$(0,0,0,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, C_1 short, C_1 open$\}$
absolute (identical residual networks)	$\{B_2$ short, C_1 short$\}$
signature (same quadruple)	$\{B_2$ short, C_1 short, C_1 open$\}$

Network 16



Network

realisation	`R p (R s L)`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$
m-word	`BBBBBcbCCB` $\rightarrow$ `BCB`
signature	$\Phi = (0,1,1,0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

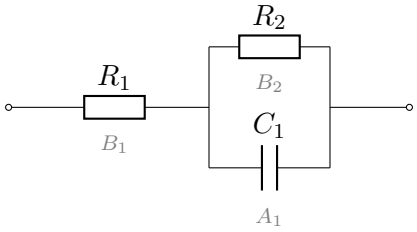
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, C_1$	`BC`	$(0,0,1,1)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1,1,0,0)$	DC, sweep	
B_2	open	B_1	`B`	$(0,0,0,0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0,0,0,0)$	sweep	
C_1	open	B_1	`B`	$(0,0,0,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, C_1 short, C_1 open}
absolute (identical residual networks)	{B_2 open, C_1 open}
signature (same quadruple)	{B_2 open, C_1 short, C_1 open}

Network 17



Network

realisation	`R s (R p C)`
m-expression	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, A_1)$
m-word	`BBBBBabAAB` $\rightarrow$ `BAB`
signature	$\Phi = (0,1,1,0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = B_1 \,\mathtt{s}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathtt{s}\, B_2$

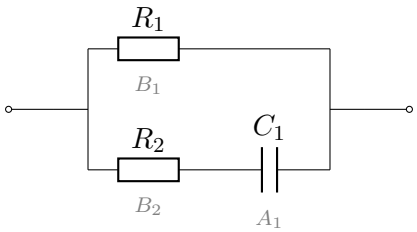
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathtt{p}\, A_1$	`BA`	$(0,1,0,-1)$	HF, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0,0,0,0)$	sweep	
B_2	open	$B_1 \,\mathtt{s}\, A_1$	`AB`	$(-1,0,1,0)$	DC, sweep	
A_1	short	B_1	`B`	$(0,0,0,0)$	sweep	
A_1	open	$B_1 \,\mathtt{s}\, B_2$	`B`	$(0,0,0,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, A_1 short, A_1 open}
absolute (identical residual networks)	{B_2 short, A_1 short}
signature (same quadruple)	{B_2 short, A_1 short, A_1 open}

Network 18



Network

realisation	`R p (R s C)`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$
m-word	`BBabABbAaBB` $\rightarrow$ `BAB`
signature	$\Phi = (0,1,1,0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 1, \quad \lambda = 2$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

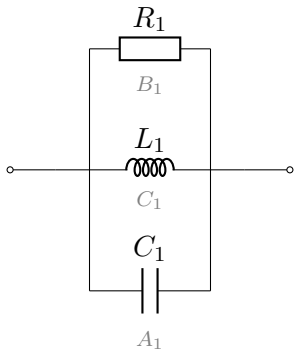
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, A_1$	`AB`	$(-1,0,1,0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0,1,0,-1)$	HF, sweep	
B_2	open	B_1	`B`	$(0,0,0,0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0,0,0,0)$	sweep	
A_1	open	B_1	`B`	$(0,0,0,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, A_1 short, A_1 open}
absolute (identical residual networks)	{B_2 open, A_1 open}
signature (same quadruple)	{B_2 open, A_1 short, A_1 open}

Network 19



Network

realisation	`R p L p C`
m-expression	B_1 p C_1 p A_1
m-word	`CCBbCcBabAA` $\rightarrow$ `CA`
signature	$\Phi = (1, 2, 0, -1)$, $Z(s) = s^1\, P/Q$, $\deg P = 0$, $\deg Q = 2$, $\lambda = 2$
corners	$\delta_0 = \mathrm{K}$, $\delta_\infty = \mathrm{K}$, $\delta_\mathrm{K} = \mathrm{K}$, $\delta_\mathrm{P} = B_1$

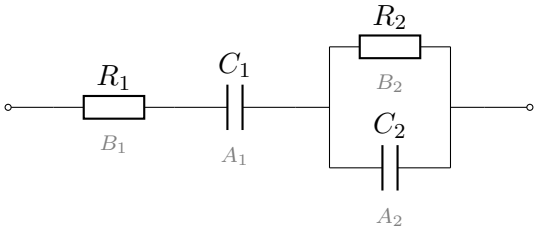
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	C_1 p A_1	`CA`	$(1, 2, 0, -1)$	none	✓
C_1	short	K	`K`	–	port	
C_1	open	B_1 p A_1	`BA`	$(0, 1, 0, -1)$	DC, sweep	
A_1	short	K	`K`	–	port	
A_1	open	B_1 p C_1	`CB`	$(1, 1, 0, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 short, C_1 short, A_1 short}
absolute (identical residual networks)	{B_1 short, C_1 short, A_1 short}
signature (same quadruple)	{*nominal*, B_1 open} {B_1 short, C_1 short, A_1 short}

Network 20



Network

realisation	`R s C s (R p C)`
m-expression	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$
m-word	`AABbAaBBBBabAAB` $\rightarrow$ `ABAB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

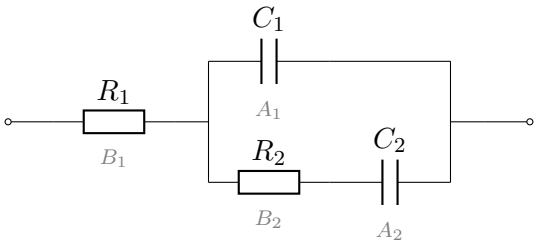
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`ABA`	$(-1, 1, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	DC, sweep	
A_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, A_2$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, B_2$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, A_1 open$\}$ $\{B_2$ short, B_2 open, A_2 short, A_2 open$\}$
absolute (identical residual networks)	$\{B_1$ open, A_1 open$\}$ $\{B_2$ short, A_2 short$\}$
signature (same quadruple)	$\{B_1$ open, A_1 open$\}$ $\{B_2$ short, B_2 open, A_2 short, A_2 open$\}$

Network 21



Network

realisation	`R s (C p (R s C))`
m-expression	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2))$
m-word	`ABbAAAAaBbAaBAB` $\rightarrow$ `ABAB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

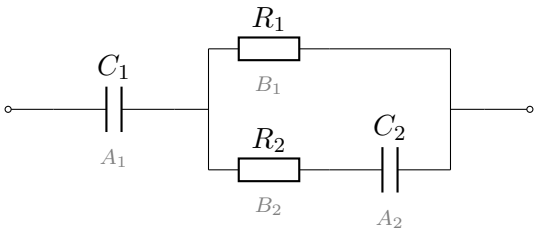
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2)$	`ABA`	$(-1, 1, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	DC, sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, A_2$	`AB`	$(-1, 0, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, A_2)$	`AB`	$(-1, 0, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	DC, sweep	
A_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{A_1 open, B_2 short, B_2 open, A_2 open}
absolute (identical residual networks)	{B_2 open, A_2 open}
signature (same quadruple)	{A_1 open, B_2 short, B_2 open, A_2 open}

Network 22



Network

realisation	`C s (R p (R s C))`
m-expression	$A_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2))$
m-word	`AAbaBBabABbAaBBB` $\rightarrow$ `ABAB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

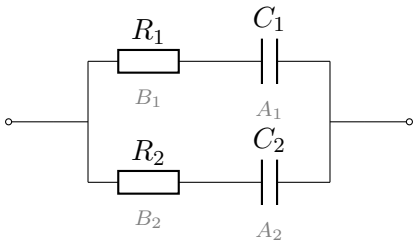
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	DC, sweep	
A_1	open	P	`P`	–	port	
B_1	short	A_1	`A`	$(-1, 0, 0, -1)$	HF, sweep	
B_1	open	$A_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, A_2$	`AB`	$(-1, 0, 1, 0)$	sweep	
B_2	short	$A_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, A_2)$	`ABA`	$(-1, 1, 1, -1)$	HF, sweep	
B_2	open	$A_1 \,\mathsf{s}\, B_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_2	short	$A_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, B_2)$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_2	open	$A_1 \,\mathsf{s}\, B_1$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 open, B_2 open, A_2 short, A_2 open}
absolute (identical residual networks)	{B_2 open, A_2 open}
signature (same quadruple)	{B_1 open, B_2 open, A_2 short, A_2 open}

Network 23



Network

realisation	`(R s C) p (R s C)`
m-expression	$(B_1 \,\mathsf{s}\, A_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2)$
m-word	`AABbAaBabABbAaBB` $\rightarrow$ `ABAB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

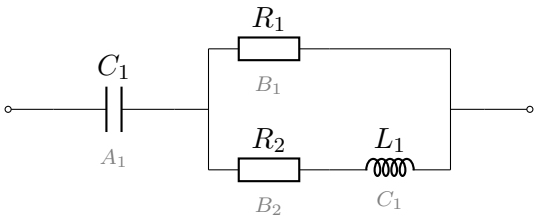
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2)$	`ABA`	$(-1, 1, 1, -1)$	HF, sweep	
B_1	open	$B_2 \,\mathsf{s}\, A_2$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	DC, sweep	
A_1	open	$B_2 \,\mathsf{s}\, A_2$	`AB`	$(-1, 0, 1, 0)$	sweep	
B_2	short	$(B_1 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2$	`ABA`	$(-1, 1, 1, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
A_2	short	$(B_1 \,\mathsf{s}\, A_1) \,\mathsf{p}\, B_2$	`BAB`	$(0, 1, 1, 0)$	DC, sweep	
A_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, B_2 short$\}$ $\{B_1$ open, A_1 open, B_2 open, A_2 open$\}$ $\{A_1$ short, A_2 short$\}$
absolute (identical residual networks)	$\{B_1$ open, A_1 open$\}$ $\{B_2$ open, A_2 open$\}$
signature (same quadruple)	$\{B_1$ short, B_2 short$\}$ $\{B_1$ open, A_1 open, B_2 open, A_2 open$\}$ $\{A_1$ short, A_2 short$\}$

Network 24



Network

realisation	`C s (R p (R s L))`
m-expression	$A_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1))$
m-word	`AAbaBBBBBcbCCBB` $\rightarrow$ `ACB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

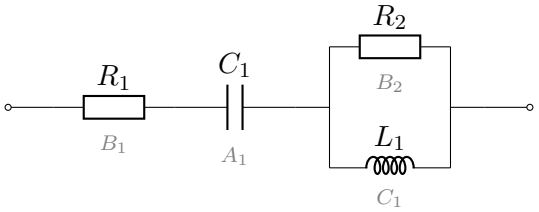
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	DC, sweep	
A_1	open	P	`P`	–	port	
B_1	short	A_1	`A`	$(-1, 0, 0, -1)$	HF, sweep	
B_1	open	$A_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, C_1$	`AC`	$(-1, 0, 2, 1)$	HF, sweep	
B_2	short	$A_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	none	✓
B_2	open	$A_1 \,\mathsf{s}\, B_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
C_1	short	$A_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, B_2)$	`AB`	$(-1, 0, 1, 0)$	sweep	
C_1	open	$A_1 \,\mathsf{s}\, B_1$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, C_1 short, C_1 open}
absolute (identical residual networks)	{B_2 open, C_1 open}
signature (same quadruple)	{*nominal*, B_2 short}
	{B_2 open, C_1 short, C_1 open}

Network 25



Network

realisation	`R s C s (R p L)`
m-expression	$B_1 \,\mathtt{s}\, A_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_1)$
m-word	`AABbAaBcbCBbCcBB` $\rightarrow$ `ACB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1 \,\mathtt{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

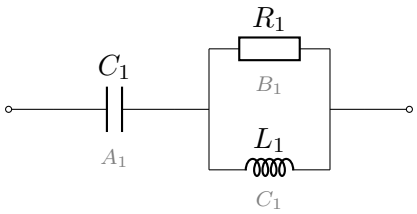
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	none	✓
B_1	open	P	`P`	–	port	
A_1	short	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	DC, sweep	
A_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathtt{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathtt{s}\, A_1 \,\mathtt{s}\, C_1$	`AC`	$(-1, 0, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathtt{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathtt{s}\, A_1 \,\mathtt{s}\, B_2$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 open, A_1 open} {B_2 short, C_1 short, C_1 open}
absolute (identical residual networks)	{B_1 open, A_1 open} {B_2 short, C_1 short}
signature (same quadruple)	{*nominal*, B_1 short} {B_1 open, A_1 open} {B_2 short, C_1 short, C_1 open}

Network 26



Network

realisation	`C s (R p L)`
m-expression	$A_1 \,\mathtt{s}\, (B_1 \,\mathtt{p}\, C_1)$
m-word	`AAcaCBbCcBB` $\rightarrow$ `ACB`
signature	$\Phi = (-1, 1, 2, 0), \quad Z(s) = s^{-1}\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{P}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

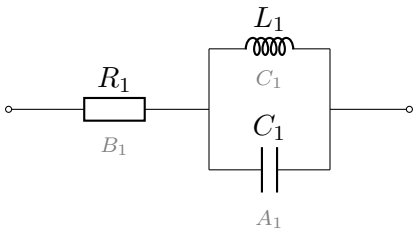
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	$B_1 \,\mathtt{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
A_1	open	P	`P`	–	port	
B_1	short	A_1	`A`	$(-1, 0, 0, -1)$	HF, sweep	
B_1	open	$A_1 \,\mathtt{s}\, C_1$	`AC`	$(-1, 0, 2, 1)$	HF, sweep	
C_1	short	A_1	`A`	$(-1, 0, 0, -1)$	HF, sweep	
C_1	open	$A_1 \,\mathtt{s}\, B_1$	`AB`	$(-1, 0, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, C_1 short$\}$
absolute (identical residual networks)	$\{B_1$ short, C_1 short$\}$
signature (same quadruple)	$\{B_1$ short, C_1 short$\}$

Network 27



Network

realisation	`R s (L p C)`
m-expression	$B_1 \,\mathtt{s}\, (C_1 \,\mathtt{p}\, A_1)$
m-word	`BBcbCCacAAB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

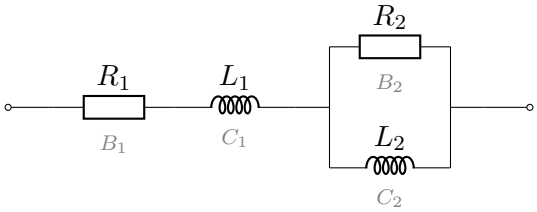
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathtt{p}\, A_1$	`CA`	$(1, 2, 0, -1)$	DC, HF, sweep	
B_1	open	P	`P`	–	port	
C_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathtt{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	$\{C_1$ short, A_1 short$\}$
absolute (identical residual networks)	$\{C_1$ short, A_1 short$\}$
signature (same quadruple)	$\{C_1$ short, A_1 short$\}$

Network 28



Network

realisation	`R s L s (R p L)`
m-expression	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_2)$
m-word	`BBBcbCCCCBbCcBC` $\rightarrow$ `BCBC`
signature	$\Phi = (0, 1, 2, 1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

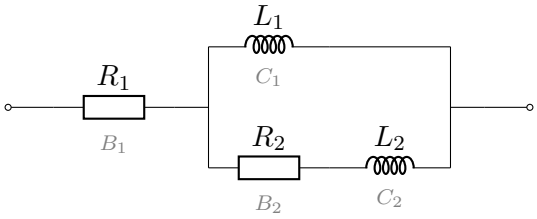
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_2)$	`CBC`	$(1, 1, 1, 1)$	DC, sweep	
B_1	open	P	`P`	–	port	
C_1	short	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	HF, sweep	
C_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	sweep	
B_2	open	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, C_2$	`BC`	$(0, 0, 1, 1)$	sweep	
C_2	short	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	sweep	
C_2	open	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, B_2$	`BC`	$(0, 0, 1, 1)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, C_1 open$\}$ $\{B_2$ short, B_2 open, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open$\}$ $\{B_2$ short, C_2 short$\}$
signature (same quadruple)	$\{B_1$ open, C_1 open$\}$ $\{B_2$ short, B_2 open, C_2 short, C_2 open$\}$

Network 29



Network

realisation	`R s (L p (R s L))`
m-expression	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2))$
m-word	`BBcbCCbcBBcbCCCC` $\rightarrow$ `BCBC`
signature	$\Phi = (0,1,2,1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

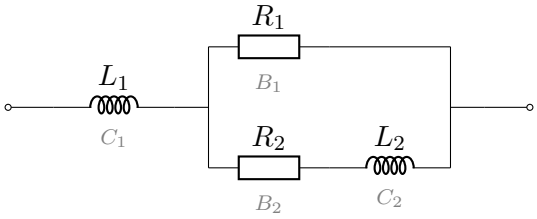
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2)$	`CBC`	$(1,1,1,1)$	DC, sweep	
B_1	open	P	`P`	–	port	
C_1	short	B_1	`B`	$(0,0,0,0)$	HF, sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, C_2$	`BC`	$(0,0,1,1)$	sweep	
B_2	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, C_2)$	`BC`	$(0,0,1,1)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0,0,1,1)$	sweep	
C_2	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_2)$	`BCB`	$(0,1,1,0)$	HF, sweep	
C_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0,0,1,1)$	sweep	

Ambiguity groups

structural (same topology after merging)	{C_1 open, B_2 short, B_2 open, C_2 open}
absolute (identical residual networks)	{B_2 open, C_2 open}
signature (same quadruple)	{C_1 open, B_2 short, B_2 open, C_2 open}

Network 30



Network

realisation	`L s (R p (R s L))`
m-expression	$C_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2))$
m-word	`BCcBBBBBbcbCCBC` $\rightarrow$ `BCBC`
signature	$\Phi = (0, 1, 2, 1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

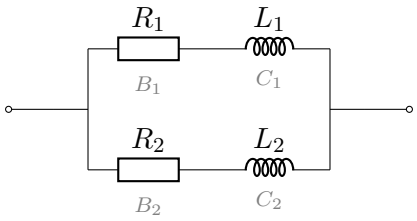
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	HF, sweep	
C_1	open	P	`P`	–	port	
B_1	short	C_1	`C`	$(1, 0, 0, 1)$	DC, sweep	
B_1	open	$C_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, C_2$	`BC`	$(0, 0, 1, 1)$	sweep	
B_2	short	$C_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, C_2)$	`CBC`	$(1, 1, 1, 1)$	DC, sweep	
B_2	open	$C_1 \,\mathsf{s}\, B_1$	`BC`	$(0, 0, 1, 1)$	sweep	
C_2	short	$C_1 \,\mathsf{s}\, (B_1 \,\mathsf{p}\, B_2)$	`BC`	$(0, 0, 1, 1)$	sweep	
C_2	open	$C_1 \,\mathsf{s}\, B_1$	`BC`	$(0, 0, 1, 1)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 open, B_2 open, C_2 short, C_2 open}
absolute (identical residual networks)	{B_2 open, C_2 open}
signature (same quadruple)	{B_1 open, B_2 open, C_2 short, C_2 open}

Network 31



Network

realisation	`(R s L) p (R s L)`
m-expression	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2)$
m-word	`BBBcbCCbcBBcbCCC` $\rightarrow$ `BCBC`
signature	$\Phi = (0, 1, 2, 1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

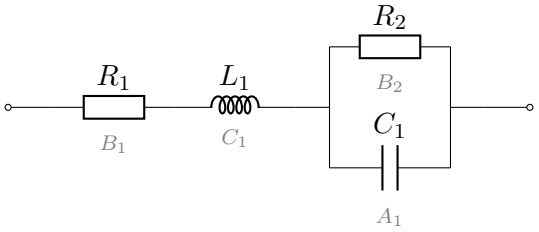
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2)$	`CBC`	$(1, 1, 1, 1)$	DC, sweep	
B_1	open	$B_2 \,\mathsf{s}\, C_2$	`BC`	$(0, 0, 1, 1)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	HF, sweep	
C_1	open	$B_2 \,\mathsf{s}\, C_2$	`BC`	$(0, 0, 1, 1)$	sweep	
B_2	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, C_2$	`CBC`	$(1, 1, 1, 1)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	sweep	
C_2	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_2$	`BCB`	$(0, 1, 1, 0)$	HF, sweep	
C_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, B_2 short$\}$ $\{B_1$ open, C_1 open, B_2 open, C_2 open$\}$ $\{C_1$ short, C_2 short$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open$\}$ $\{B_2$ open, C_2 open$\}$
signature (same quadruple)	$\{B_1$ short, B_2 short$\}$ $\{B_1$ open, C_1 open, B_2 open, C_2 open$\}$ $\{C_1$ short, C_2 short$\}$

Network 32



Network

realisation	`R s L s (R p C)`
m-expression	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, A_1)$
m-word	`BBBcbCCcBBbabAAC` $\rightarrow$ `BAC`
signature	$\Phi = (0, 1, 2, 1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1 \,\mathtt{s}\, B_2, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

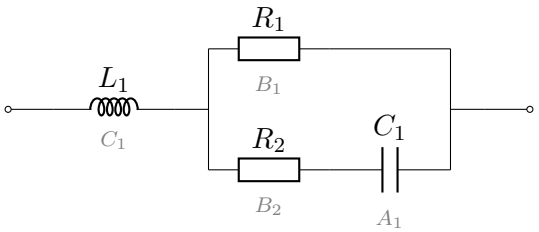
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, A_1)$	`BAC`	$(0, 1, 2, 1)$	none	✓
B_1	open	P	`P`	–	port	
C_1	short	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	
C_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	sweep	
B_2	open	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, A_1$	`AC`	$(-1, 0, 2, 1)$	DC, sweep	
A_1	short	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	sweep	
A_1	open	$B_1 \,\mathtt{s}\, C_1 \,\mathtt{s}\, B_2$	`BC`	$(0, 0, 1, 1)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, C_1 open$\}$ $\{B_2$ short, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open$\}$ $\{B_2$ short, A_1 short$\}$
signature (same quadruple)	$\{$*nominal*, B_1 short$\}$ $\{B_1$ open, C_1 open$\}$ $\{B_2$ short, A_1 short, A_1 open$\}$

Network 33



Network

realisation	`L s (R p (R s C))`
m-expression	$C_1 \,\mathtt{s}\, (B_1 \,\mathtt{p}\, (B_2 \,\mathtt{s}\, A_1))$
m-word	`BCcBBbabABbAaBBC` $\rightarrow$ `BAC`
signature	$\Phi = (0, 1, 2, 1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = B_1 \,\mathtt{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

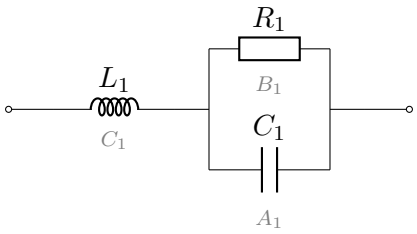
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	$B_1 \,\mathtt{p}\, (B_2 \,\mathtt{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	
C_1	open	P	`P`	–	port	
B_1	short	C_1	`C`	$(1, 0, 0, 1)$	DC, sweep	
B_1	open	$C_1 \,\mathtt{s}\, B_2 \,\mathtt{s}\, A_1$	`AC`	$(-1, 0, 2, 1)$	DC, sweep	
B_2	short	$C_1 \,\mathtt{s}\, (B_1 \,\mathtt{p}\, A_1)$	`BAC`	$(0, 1, 2, 1)$	none	✓
B_2	open	$C_1 \,\mathtt{s}\, B_1$	`BC`	$(0, 0, 1, 1)$	sweep	
A_1	short	$C_1 \,\mathtt{s}\, (B_1 \,\mathtt{p}\, B_2)$	`BC`	$(0, 0, 1, 1)$	sweep	
A_1	open	$C_1 \,\mathtt{s}\, B_1$	`BC`	$(0, 0, 1, 1)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, A_1 short, A_1 open}
absolute (identical residual networks)	{B_2 open, A_1 open}
signature (same quadruple)	{*nominal*, B_2 short}
	{B_2 open, A_1 short, A_1 open}

Network 34



Network

realisation	`L s (R p C)`
m-expression	$C_1 \,\mathtt{s}\, (B_1 \,\mathtt{p}\, A_1)$
m-word	`BCcBBbabAAC` $\rightarrow$ `BAC`
signature	$\Phi = (0, 1, 2, 1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 1, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{P}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

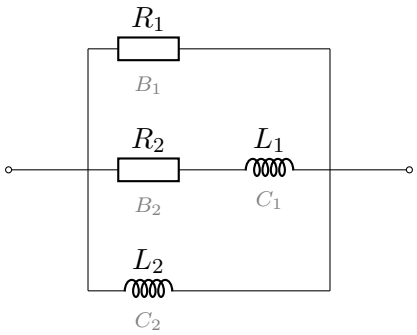
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	$B_1 \,\mathtt{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
C_1	open	P	`P`	–	port	
B_1	short	C_1	`C`	$(1, 0, 0, 1)$	DC, sweep	
B_1	open	$C_1 \,\mathtt{s}\, A_1$	`AC`	$(-1, 0, 2, 1)$	DC, sweep	
A_1	short	C_1	`C`	$(1, 0, 0, 1)$	DC, sweep	
A_1	open	$C_1 \,\mathtt{s}\, B_1$	`BC`	$(0, 0, 1, 1)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_1 short$\}$
absolute (identical residual networks)	$\{B_1$ short, A_1 short$\}$
signature (same quadruple)	$\{B_1$ short, A_1 short$\}$

Network 35



Network

realisation	`R p (R s L) p L`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, C_2$
m-word	`CBBBBBcbCCBbCcB` $\rightarrow$ `CBCB`
signature	$\Phi = (1,2,1,0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

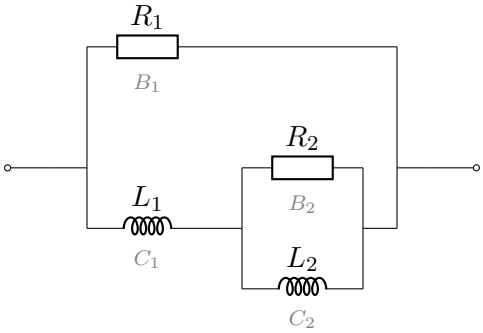
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, C_2$	`CBC`	$(1,1,1,1)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, C_2$	`CB`	$(1,1,0,0)$	sweep	
B_2	open	$B_1 \,\mathsf{p}\, C_2$	`CB`	$(1,1,0,0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, C_2$	`CB`	$(1,1,0,0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, C_2$	`CB`	$(1,1,0,0)$	sweep	
C_2	short	K	`K`	–	port	
C_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0,1,1,0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, C_2 short$\}$ $\{B_2$ short, B_2 open, C_1 short, C_1 open$\}$
absolute (identical residual networks)	$\{B_1$ short, C_2 short$\}$ $\{B_2$ open, C_1 open$\}$
signature (same quadruple)	$\{B_1$ short, C_2 short$\}$ $\{B_2$ short, B_2 open, C_1 short, C_1 open$\}$

Network 36



Network

realisation	`R p (L s (R p L))`
m-expression	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2))$
m-word	`CBbCCCCcBbCcBCB` $\rightarrow$ `CBCB`
signature	$\Phi = (1,2,1,0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

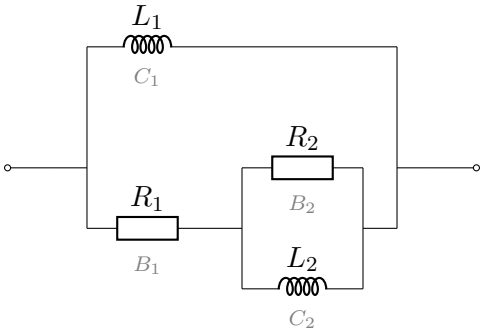
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2)$	`CBC`	$(1,1,1,1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, C_2$	`CB`	$(1,1,0,0)$	sweep	
C_1	open	B_1	`B`	$(0,0,0,0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1,1,0,0)$	sweep	
B_2	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, C_2)$	`CB`	$(1,1,0,0)$	sweep	
C_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1,1,0,0)$	sweep	
C_2	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0,1,1,0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{C_1$ short, B_2 short, B_2 open, C_2 short$\}$
absolute (identical residual networks)	$\{B_2$ short, C_2 short$\}$
signature (same quadruple)	$\{C_1$ short, B_2 short, B_2 open, C_2 short$\}$

Network 37



Network

realisation	`L p (R s (R p L))`
m-expression	$C_1 \,\mathsf{p}\, (B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2))$
m-word	`CCbcBBcbCBbCcBBB` $\rightarrow$ `CBCB`
signature	$\Phi = (1,2,1,0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

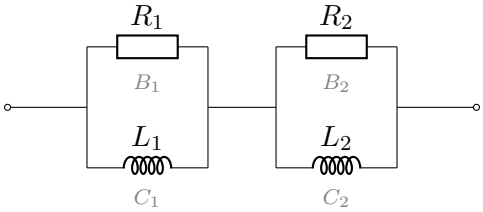
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	K	`K`	–	port	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2)$	`BCB`	$(0,1,1,0)$	DC, sweep	
B_1	short	$C_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, C_2$	`CB`	$(1,1,0,0)$	sweep	
B_1	open	C_1	`C`	$(1,0,0,1)$	HF, sweep	
B_2	short	$C_1 \,\mathsf{p}\, B_1$	`CB`	$(1,1,0,0)$	sweep	
B_2	open	$C_1 \,\mathsf{p}\, (B_1 \,\mathsf{s}\, C_2)$	`CBC`	$(1,1,1,1)$	HF, sweep	
C_2	short	$C_1 \,\mathsf{p}\, B_1$	`CB`	$(1,1,0,0)$	sweep	
C_2	open	$C_1 \,\mathsf{p}\, (B_1 \,\mathsf{s}\, B_2)$	`CB`	$(1,1,0,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, B_2 short, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_2$ short, C_2 short$\}$
signature (same quadruple)	$\{B_1$ short, B_2 short, C_2 short, C_2 open$\}$

Network 38



Network

realisation	`(R p L) s (R p L)`
m-expression	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2)$
m-word	`CCBbCcBcbCBbCcBB` $\rightarrow$ `CBCB`
signature	$\Phi = (1, 2, 1, 0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

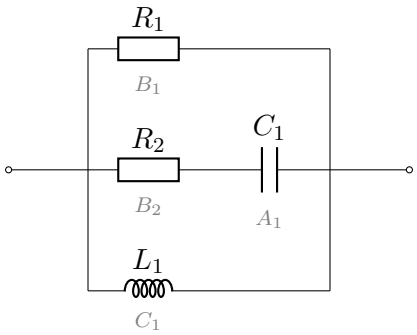
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, C_2$	`CB`	$(1, 1, 0, 0)$	sweep	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2)$	`CBC`	$(1, 1, 1, 1)$	HF, sweep	
C_1	short	$B_2 \,\mathsf{p}\, C_2$	`CB`	$(1, 1, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	sweep	
B_2	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, C_2$	`CBC`	$(1, 1, 1, 1)$	HF, sweep	
C_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	sweep	
C_2	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_2$	`BCB`	$(0, 1, 1, 0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, C_1 short, B_2 short, C_2 short$\}$ $\{B_1$ open, B_2 open$\}$ $\{C_1$ open, C_2 open$\}$
absolute (identical residual networks)	$\{B_1$ short, C_1 short$\}$ $\{B_2$ short, C_2 short$\}$
signature (same quadruple)	$\{B_1$ short, C_1 short, B_2 short, C_2 short$\}$ $\{B_1$ open, B_2 open$\}$ $\{C_1$ open, C_2 open$\}$

Network 39



Network

realisation	`R p (R s C) p L`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, C_1$
m-word	`CBBabABbAaBBbCcB` $\rightarrow$ `CAB`
signature	$\Phi = (1, 2, 1, 0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

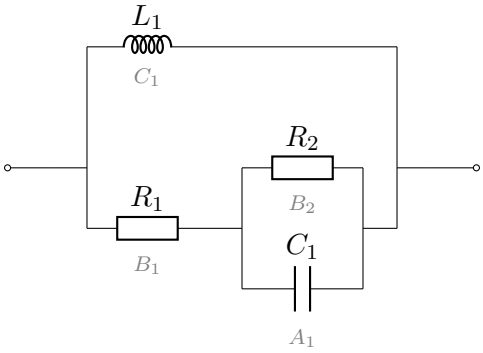
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, C_1$	`CAB`	$(1, 2, 1, 0)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, A_1 \,\mathsf{p}\, C_1$	`CA`	$(1, 2, 0, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	sweep	
C_1	short	K	`K`	–	port	
C_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, C_1 short$\}$ $\{B_2$ open, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ short, C_1 short$\}$ $\{B_2$ open, A_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_1 open$\}$ $\{B_1$ short, C_1 short$\}$ $\{B_2$ open, A_1 short, A_1 open$\}$

Network 40



Network

realisation	`L p (R s (R p C))`
m-expression	$C_1\,\mathsf{p}\,(B_1\,\mathsf{s}\,(B_2\,\mathsf{p}\,A_1))$
m-word	`CCbcBBBBBabAABB` $\rightarrow$ `CAB`
signature	$\Phi = (1,2,1,0), \quad Z(s) = s^1\,P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1\,\mathsf{s}\,B_2$

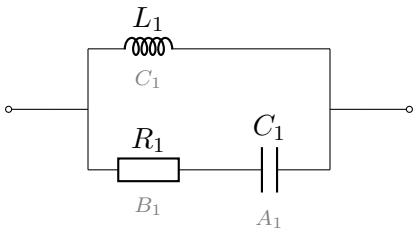
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	K	`K`	–	port	
C_1	open	$B_1\,\mathsf{s}\,(B_2\,\mathsf{p}\,A_1)$	`BAB`	$(0,1,1,0)$	DC, sweep	
B_1	short	$C_1\,\mathsf{p}\,B_2\,\mathsf{p}\,A_1$	`CA`	$(1,2,0,-1)$	HF, sweep	
B_1	open	C_1	`C`	$(1,0,0,1)$	HF, sweep	
B_2	short	$C_1\,\mathsf{p}\,B_1$	`CB`	$(1,1,0,0)$	sweep	
B_2	open	$C_1\,\mathsf{p}\,(B_1\,\mathsf{s}\,A_1)$	`CAB`	$(1,2,1,0)$	none	✓
A_1	short	$C_1\,\mathsf{p}\,B_1$	`CB`	$(1,1,0,0)$	sweep	
A_1	open	$C_1\,\mathsf{p}\,(B_1\,\mathsf{s}\,B_2)$	`CB`	$(1,1,0,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, A_1 short, A_1 open}
absolute (identical residual networks)	{B_2 short, A_1 short}
signature (same quadruple)	{*nominal*, B_2 open} {B_2 short, A_1 short, A_1 open}

Network 41



Network

realisation	`L p (R s C)`
m-expression	$C_1 \,\mathtt{p}\, (B_1 \,\mathtt{s}\, A_1)$
m-word	`CCacABbAaBB` $\rightarrow$ `CAB`
signature	$\Phi = (1, 2, 1, 0), \quad Z(s) = s^1\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = \mathrm{K}, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

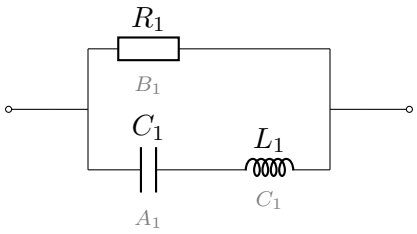
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
C_1	short	K	`K`	–	port	
C_1	open	$B_1 \,\mathtt{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_1	short	$C_1 \,\mathtt{p}\, A_1$	`CA`	$(1, 2, 0, -1)$	HF, sweep	
B_1	open	C_1	`C`	$(1, 0, 0, 1)$	HF, sweep	
A_1	short	$C_1 \,\mathtt{p}\, B_1$	`CB`	$(1, 1, 0, 0)$	sweep	
A_1	open	C_1	`C`	$(1, 0, 0, 1)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, A_1 open$\}$
signature (same quadruple)	$\{B_1$ open, A_1 open$\}$

Network 42



Network

realisation	`R p (C s L)`
m-expression	$B_1 \,\mathtt{p}\, (A_1 \,\mathtt{s}\, C_1)$
m-word	`BBabAAcaCCB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

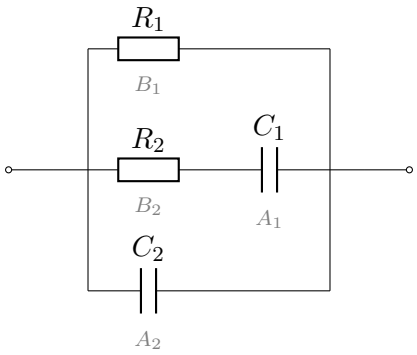
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$A_1 \,\mathtt{s}\, C_1$	`AC`	$(-1, 0, 2, 1)$	DC, HF, sweep	
A_1	short	$B_1 \,\mathtt{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	short	$B_1 \,\mathtt{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{A_1$ open, C_1 open$\}$
absolute (identical residual networks)	$\{A_1$ open, C_1 open$\}$
signature (same quadruple)	$\{A_1$ open, C_1 open$\}$

Network 43



Network

realisation	`R p (R s C) p C`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2$
m-word	`BBBabABbAaBBabAA` $\rightarrow$ `BABA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

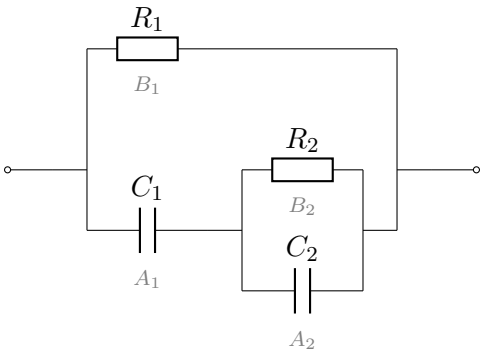
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2$	`ABA`	$(-1, 1, 1, -1)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
B_2	open	$B_1 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
A_2	short	K	`K`	–	port	
A_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_2 short$\}$ $\{B_2$ short, B_2 open, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ short, A_2 short$\}$ $\{B_2$ open, A_1 open$\}$
signature (same quadruple)	$\{B_1$ short, A_2 short$\}$ $\{B_2$ short, B_2 open, A_1 short, A_1 open$\}$

Network 44



Network

realisation	`R p (C s (R p C))`
m-expression	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2))$
m-word	`BBabAAbaBBabAAAA` $\rightarrow$ `BABA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

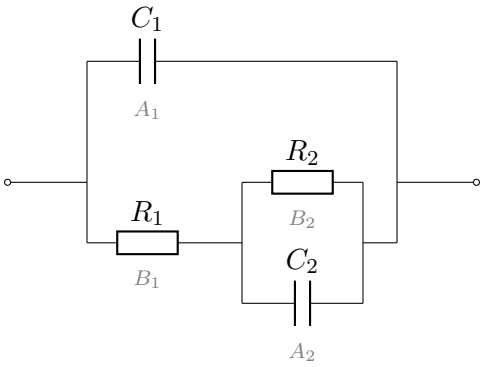
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`ABA`	$(-1, 1, 1, -1)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
B_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, A_2)$	`BA`	$(0, 1, 0, -1)$	sweep	
A_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
A_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{A_1 short, B_2 short, B_2 open, A_2 short}
absolute (identical residual networks)	{B_2 short, A_2 short}
signature (same quadruple)	{A_1 short, B_2 short, B_2 open, A_2 short}

Network 45



Network

realisation	`C p (R s (R p C))`
m-expression	$A_1\ \mathsf{p}\ (B_1\ \mathsf{s}\ (B_2\ \mathsf{p}\ A_2))$
m-word	`BAaBBBBBbabAABA` $\rightarrow$ `BABA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1\ \mathsf{s}\ B_2, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1\ \mathsf{s}\ B_2$

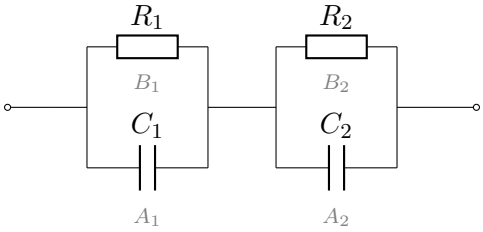
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	K	`K`	–	port	
A_1	open	$B_1\ \mathsf{s}\ (B_2\ \mathsf{p}\ A_2)$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	
B_1	short	$A_1\ \mathsf{p}\ B_2\ \mathsf{p}\ A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
B_1	open	A_1	`A`	$(-1, 0, 0, -1)$	DC, sweep	
B_2	short	$A_1\ \mathsf{p}\ B_1$	`BA`	$(0, 1, 0, -1)$	sweep	
B_2	open	$A_1\ \mathsf{p}\ (B_1\ \mathsf{s}\ A_2)$	`ABA`	$(-1, 1, 1, -1)$	DC, sweep	
A_2	short	$A_1\ \mathsf{p}\ B_1$	`BA`	$(0, 1, 0, -1)$	sweep	
A_2	open	$A_1\ \mathsf{p}\ (B_1\ \mathsf{s}\ B_2)$	`BA`	$(0, 1, 0, -1)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, B_2 short, A_2 short, A_2 open$\}$
absolute (identical residual networks)	$\{B_2$ short, A_2 short$\}$
signature (same quadruple)	$\{B_1$ short, B_2 short, A_2 short, A_2 open$\}$

Network 46



Network

realisation	`(R p C) s (R p C)`
m-expression	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$
m-word	`BBBabAAbaBBabAAA` $\rightarrow$ `BABA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

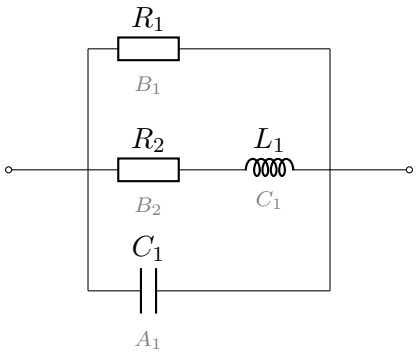
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`ABA`	$(-1, 1, 1, -1)$	DC, sweep	
A_1	short	$B_2 \,\mathsf{p}\, A_2$	`BA`	$(0, 1, 0, -1)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
B_2	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, A_2$	`ABA`	$(-1, 1, 1, -1)$	DC, sweep	
A_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
A_2	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_2$	`BAB`	$(0, 1, 1, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_1 short, B_2 short, A_2 short$\}$ $\{B_1$ open, B_2 open$\}$ $\{A_1$ open, A_2 open$\}$
absolute (identical residual networks)	$\{B_1$ short, A_1 short$\}$ $\{B_2$ short, A_2 short$\}$
signature (same quadruple)	$\{B_1$ short, A_1 short, B_2 short, A_2 short$\}$ $\{B_1$ open, B_2 open$\}$ $\{A_1$ open, A_2 open$\}$

Network 47



Network

realisation	`R p (R s L) p C`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1$
m-word	`BBBBBBcbCCBabAA` $\rightarrow$ `BCA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

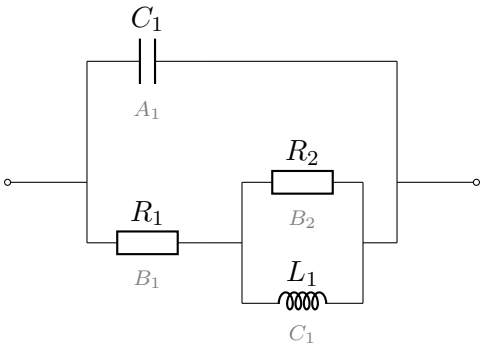
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1$	`BCA`	$(0, 2, 1, -1)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, A_1$	`CA`	$(1, 2, 0, -1)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
A_1	short	K	`K`	–	port	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_1 short$\}$ $\{B_2$ open, C_1 short, C_1 open$\}$
absolute (identical residual networks)	$\{B_1$ short, A_1 short$\}$ $\{B_2$ open, C_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_1 open$\}$ $\{B_1$ short, A_1 short$\}$ $\{B_2$ open, C_1 short, C_1 open$\}$

Network 48



Network

realisation	`C p (R s (R p L))`
m-expression	$A_1 \,\mathsf{p}\, (B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1))$
m-word	`BAaBBbcbCBbCcBBA` $\rightarrow$ `BCA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

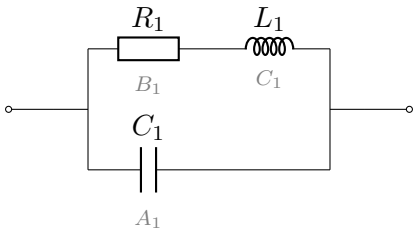
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	K	`K`	–	port	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	HF, sweep	
B_1	short	$A_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, C_1$	`CA`	$(1, 2, 0, -1)$	DC, sweep	
B_1	open	A_1	`A`	$(-1, 0, 0, -1)$	DC, sweep	
B_2	short	$A_1 \,\mathsf{p}\, B_1$	`BA`	$(0, 1, 0, -1)$	sweep	
B_2	open	$A_1 \,\mathsf{p}\, (B_1 \,\mathsf{s}\, C_1)$	`BCA`	$(0, 2, 1, -1)$	none	✓
C_1	short	$A_1 \,\mathsf{p}\, B_1$	`BA`	$(0, 1, 0, -1)$	sweep	
C_1	open	$A_1 \,\mathsf{p}\, (B_1 \,\mathsf{s}\, B_2)$	`BA`	$(0, 1, 0, -1)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, C_1 short, C_1 open}
absolute (identical residual networks)	{B_2 short, C_1 short}
signature (same quadruple)	{*nominal*, B_2 open} {B_2 short, C_1 short, C_1 open}

Network 49



Network

realisation	`(R s L) p C`
m-expression	$(B_1 \,\mathtt{s}\, C_1) \,\mathtt{p}\, A_1$
m-word	`BBBcbCCacAA` $\rightarrow$ `BCA`
signature	$\Phi = (0, 2, 1, -1), \quad Z(s) = s^0\, P/Q, \quad \deg P = 1, \quad \deg Q = 2, \quad \lambda = 3$
corners	$\delta_0 = B_1, \quad \delta_\infty = \mathrm{K}, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = \mathrm{P}$

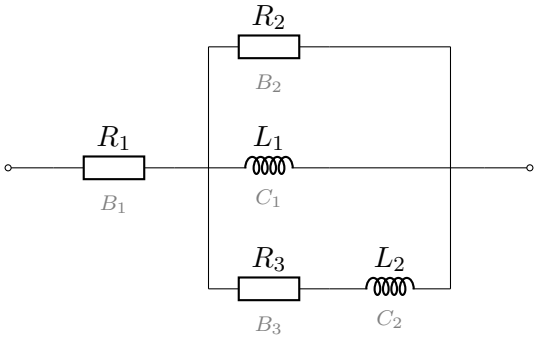
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathtt{p}\, A_1$	`CA`	$(1, 2, 0, -1)$	DC, sweep	
B_1	open	A_1	`A`	$(-1, 0, 0, -1)$	DC, sweep	
C_1	short	$B_1 \,\mathtt{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	sweep	
C_1	open	A_1	`A`	$(-1, 0, 0, -1)$	DC, sweep	
A_1	short	K	`K`	–	port	
A_1	open	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 open, C_1 open}
absolute (identical residual networks)	{B_1 open, C_1 open}
signature (same quadruple)	{B_1 open, C_1 open}

Network 50



Network

realisation	`R s (R p L p (R s L))`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$
m-word	`BBcbCCBbCcBBBBcbCCBB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

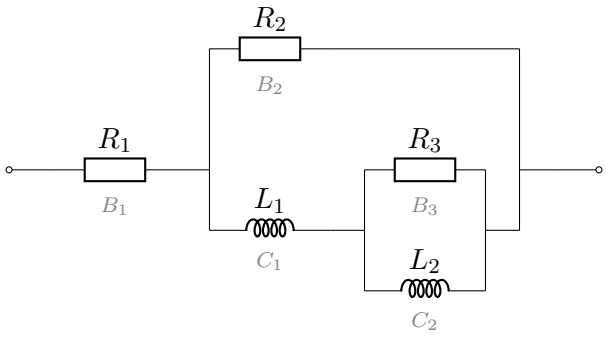
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1 \,\mathsf{p}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, C_1 short$\}$ $\{B_3$ short, B_3 open, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_2$ short, C_1 short$\}$ $\{B_3$ open, C_2 open$\}$
signature (same quadruple)	$\{B_2$ short, C_1 short$\}$ $\{C_1$ open, B_3 short, B_3 open, C_2 short, C_2 open$\}$

Network 51



Network

realisation	`R s (R p (L s (R p L)))`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)))$
m-word	`BBcbCBbCCCCcBbCcBCBB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0,2,2,0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

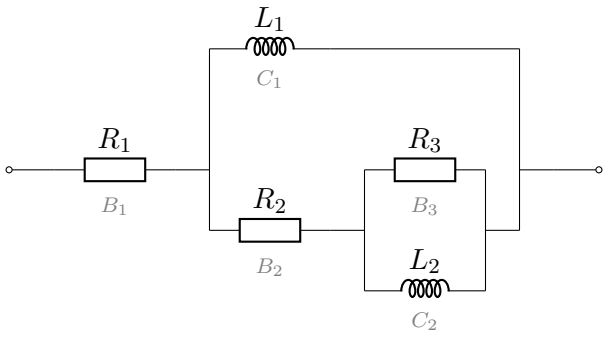
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$	`CBCB`	$(1,2,1,0)$	DC, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0,0,0,0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCBC`	$(0,1,2,1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0,1,1,0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2$	`B`	$(0,0,0,0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0,1,1,0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, C_2))$	`BCB`	$(0,1,1,0)$	sweep	
C_2	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0,1,1,0)$	sweep	
C_2	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_3))$	`BCB`	$(0,1,1,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, C_1 open$\}$ $\{C_1$ short, B_3 short, B_3 open, C_2 short$\}$
absolute (identical residual networks)	$\{B_3$ short, C_2 short$\}$
signature (same quadruple)	$\{B_2$ short, C_1 open$\}$ $\{C_1$ short, B_3 short, B_3 open, C_2 short, C_2 open$\}$

Network 52



Network

realisation	`R s (L p (R s (R p L)))`
m-expression	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)))$
m-word	`BBcbCCbcBBcbCBbCcBBBB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0,2,2,0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

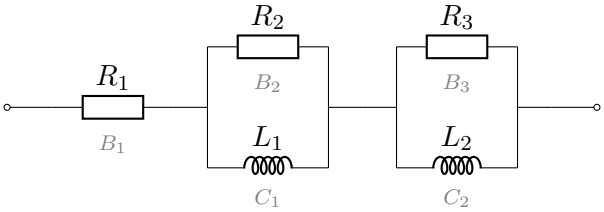
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$	`CBCB`	$(1,2,1,0)$	DC, sweep	
B_1	open	P	`P`	–	port	
C_1	short	B_1	`B`	$(0,0,0,0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0,1,1,0)$	sweep	
B_2	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0,1,1,0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0,0,1,1)$	HF, sweep	
B_3	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_2)$	`BCB`	$(0,1,1,0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2))$	`BCBC`	$(0,1,2,1)$	HF, sweep	
C_2	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_2)$	`BCB`	$(0,1,1,0)$	sweep	
C_2	open	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3))$	`BCB`	$(0,1,1,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{C_1 open, B_2 short, B_3 short, C_2 short, C_2 open}
absolute (identical residual networks)	{B_3 short, C_2 short}
signature (same quadruple)	{C_1 open, B_2 short, B_3 short, C_2 short, C_2 open}

Network 53



Network

realisation	`R s (R p L) s (R p L)`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$
m-word	`BBBcbCBbCcBBcbCBbCcBB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

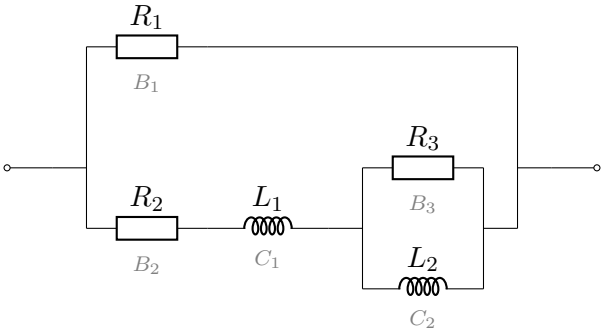
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$(B_2 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1) \,\mathsf{s}\, C_2$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
C_2	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_3$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, C_1 short, C_1 open, B_3 short, C_2 short, C_2 open$\}$ $\{B_2$ open, B_3 open$\}$
absolute (identical residual networks)	$\{B_2$ short, C_1 short$\}$ $\{B_3$ short, C_2 short$\}$
signature (same quadruple)	$\{B_2$ short, C_1 short, C_1 open, B_3 short, C_2 short, C_2 open$\}$ $\{B_2$ open, B_3 open$\}$

Network 54



Network

realisation	`R p (R s L s (R p L))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$
m-word	`BBBBBBcbCCCCBbCcBCB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

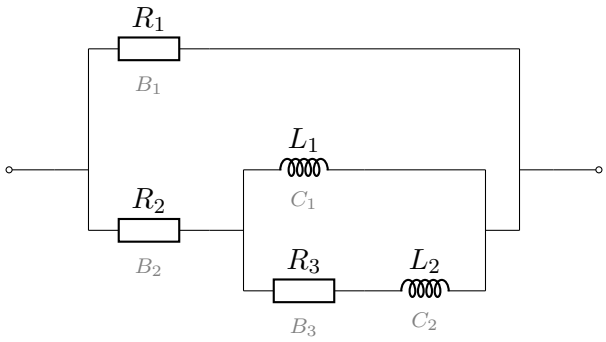
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1 \,\mathsf{s}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, C_1 open$\}$ $\{B_3$ short, B_3 open, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_2$ open, C_1 open$\}$ $\{B_3$ short, C_2 short$\}$
signature (same quadruple)	$\{B_2$ open, C_1 open$\}$ $\{C_1$ short, B_3 short, B_3 open, C_2 short, C_2 open$\}$

Network 55



Network

realisation	`R p (R s (L p (R s L)))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)))$
m-word	`BBBBBcbCCbcBBcbCCCCB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

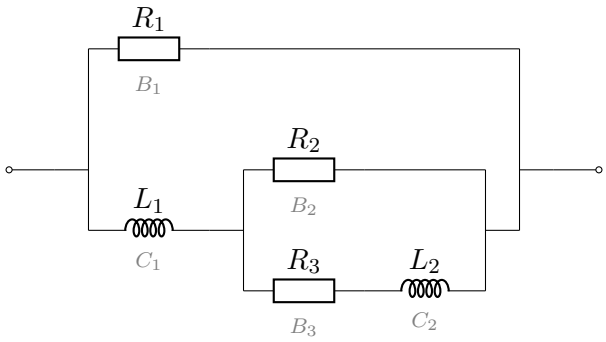
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, C_2))$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_3))$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, C_1 short$\}$ $\{C_1$ open, B_3 short, B_3 open, C_2 open$\}$
absolute (identical residual networks)	$\{B_3$ open, C_2 open$\}$
signature (same quadruple)	$\{B_2$ open, C_1 short$\}$ $\{C_1$ open, B_3 short, B_3 open, C_2 short, C_2 open$\}$

Network 56



Network

realisation	`R p (L s (R p (R s L)))`
m-expression	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)))$
m-word	`BBBBCcBBBBBbcbCCBCB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1$

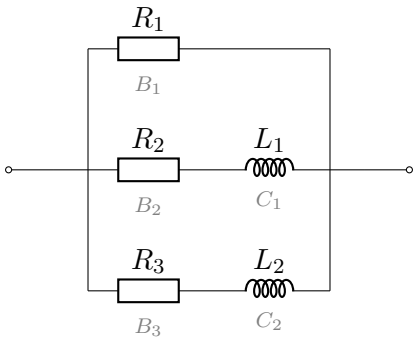
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2))$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_3	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3))$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{C_1 short, B_2 open, B_3 open, C_2 short, C_2 open}
absolute (identical residual networks)	{B_3 open, C_2 open}
signature (same quadruple)	{C_1 short, B_2 open, B_3 open, C_2 short, C_2 open}

Network 57



Network

realisation	`R p (R s L) p (R s L)`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$
m-word	`BBBBBBcbCCBBBBcbCCB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1$

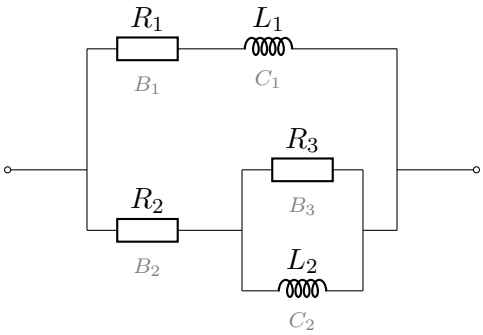
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, C_2$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_3$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, B_3 short$\}$ $\{B_2$ open, C_1 short, C_1 open, B_3 open, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_2$ open, C_1 open$\}$ $\{B_3$ open, C_2 open$\}$
signature (same quadruple)	$\{B_2$ short, B_3 short$\}$ $\{B_2$ open, C_1 short, C_1 open, B_3 open, C_2 short, C_2 open$\}$

Network 58



Network

realisation	`(R s L) p (R s (R p L))`
m-expression	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$
m-word	`BBBcbCCbcBBcbCBbCcBBB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_2 \,\mathsf{s}\, B_3, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_2 \,\mathsf{s}\, B_3$

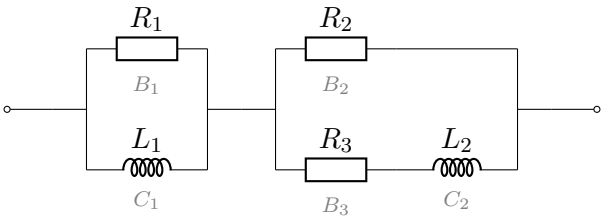
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_1	open	$B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2))$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_2$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
B_3	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_2)$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
C_2	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, C_1 open$\}$ $\{B_3$ short, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open$\}$ $\{B_3$ short, C_2 short$\}$
signature (same quadruple)	$\{B_1$ short, B_2 short$\}$ $\{B_1$ open, C_1 short, C_1 open, B_3 short, C_2 short, C_2 open$\}$

Network 59



Network

realisation	`(R p L) s (R p (R s L))`
m-expression	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$
m-word	`BCBbCcBBBBBBBcbCCBB` $\rightarrow$ `BCBCB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_2 \,\mathsf{p}\, B_3, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

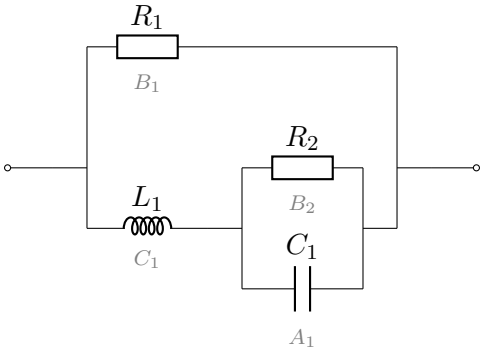
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_2))$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_2	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_3 \,\mathsf{s}\, C_2$	`BCBC`	$(0, 1, 2, 1)$	HF, sweep	
B_3	short	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_2)$	`CBCB`	$(1, 2, 1, 0)$	DC, sweep	
B_3	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	short	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_2	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, C_1 short$\}$ $\{B_3$ open, C_2 short, C_2 open$\}$
absolute (identical residual networks)	$\{B_1$ short, C_1 short$\}$ $\{B_3$ open, C_2 open$\}$
signature (same quadruple)	$\{B_1$ short, C_1 short, C_1 open, B_3 open, C_2 short, C_2 open$\}$ $\{B_1$ open, B_2 open$\}$

Network 62



Network

realisation	`R p (L s (R p C))`
m-expression	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1))$
m-word	`BBBBCcBBbabAACB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1$

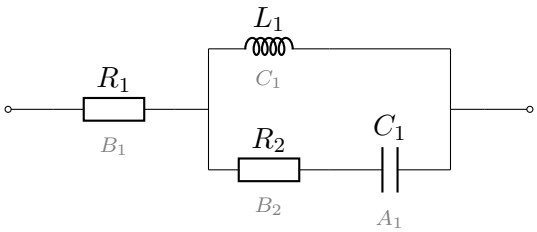
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, A_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
A_1	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
A_1	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, A_1 short}
absolute (identical residual networks)	{B_2 short, A_1 short}
signature (same quadruple)	{*nominal*, B_2 open} {B_2 short, A_1 short}

Network 63



Network

realisation	`R s (L p (R s C))`
m-expression	$B_1 \,\mathtt{s}\, (C_1 \,\mathtt{p}\, (B_2 \,\mathtt{s}\, A_1))$
m-word	`BBcbCCacABbAaBBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathtt{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

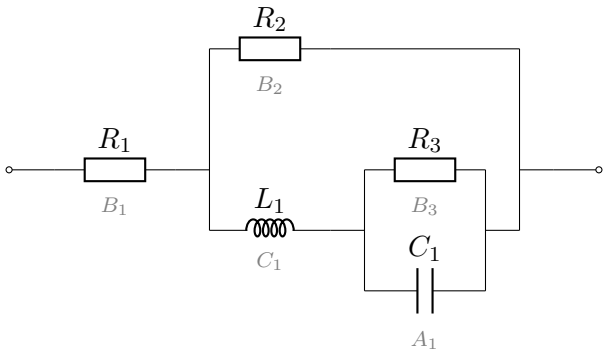
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathtt{p}\, (B_2 \,\mathtt{s}\, A_1)$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_1	open	P	`P`	–	port	
C_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathtt{s}\, B_2 \,\mathtt{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_2	short	$B_1 \,\mathtt{s}\, (C_1 \,\mathtt{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	open	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
A_1	short	$B_1 \,\mathtt{s}\, (C_1 \,\mathtt{p}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathtt{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, A_1 open}
absolute (identical residual networks)	{B_2 open, A_1 open}
signature (same quadruple)	{*nominal*, B_2 short} {B_2 open, A_1 open}

Network 64



Network

realisation	`R s (R p (L s (R p C)))`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)))$
m-word	`BBBBBBBCcBBbabAACBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3), \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

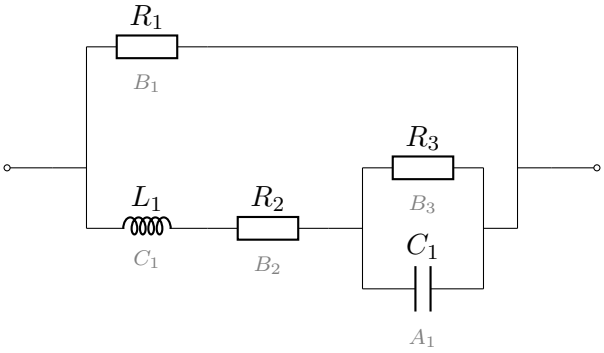
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, A_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_3))$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, C_1 open} {B_3 short, A_1 short}
absolute (identical residual networks)	{B_3 short, A_1 short}
signature (same quadruple)	{*nominal*, B_1 short, B_3 open} {B_2 short, C_1 open} {C_1 short, B_3 short, A_1 short, A_1 open}

Network 65



Network

realisation	`R p (L s R s (R p C))`
m-expression	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$
m-word	`BBBBBCcBbCcBBbabAACB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$, $\delta_\infty = B_1$, $\delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2$, $\delta_\mathrm{P} = B_1$

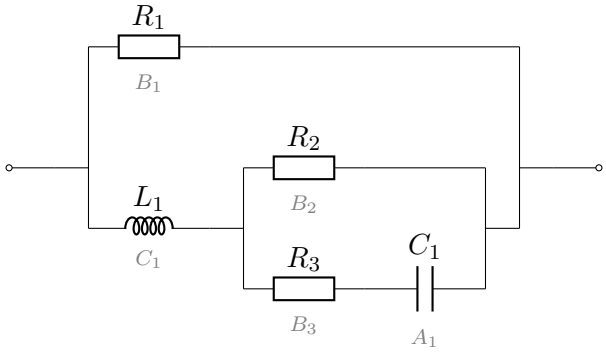
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$C_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, A_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
A_1	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{C_1 open, B_2 open} {B_3 short, A_1 short, A_1 open}
absolute (identical residual networks)	{C_1 open, B_2 open} {B_3 short, A_1 short}
signature (same quadruple)	{*nominal*, B_2 short, B_3 open} {C_1 short, B_3 short, A_1 short, A_1 open} {C_1 open, B_2 open}

Network 66



Network

realisation	`R p (L s (R p (R s C)))`
m-expression	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)))$
m-word	`BBBBCcBBbabABbAaBBCB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1$

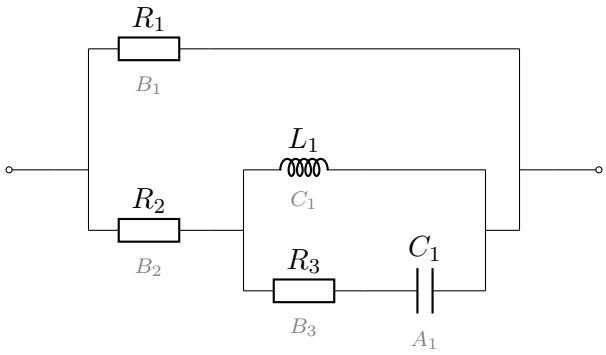
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_3	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3))$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, B_2)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_3 open, A_1 short, A_1 open}
absolute (identical residual networks)	{B_3 open, A_1 open}
signature (same quadruple)	{*nominal*, B_2 open, B_3 short} {C_1 short, B_3 open, A_1 short, A_1 open}

Network 67



Network

realisation	`R p (R s (L p (R s C)))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)))$
m-word	`BBBBBcbCCacABbAaBBBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $\quad Z(s) = s^0\, P/Q$, $\quad \deg P = 2$, $\quad \deg Q = 2$, $\quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2$, $\quad \delta_\infty = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$, $\quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2$, $\quad \delta_\mathrm{P} = B_1$

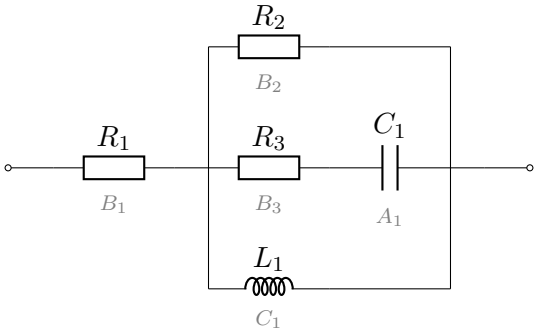
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, A_1))$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_3))$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, C_1 short$\}$ $\{B_3$ open, A_1 open$\}$
absolute (identical residual networks)	$\{B_3$ open, A_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_1 open, B_3 short$\}$ $\{B_2$ open, C_1 short$\}$ $\{C_1$ open, B_3 open, A_1 short, A_1 open$\}$

Network 68



Network

realisation	`R s (R p (R s C) p L)`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1) \,\mathsf{p}\, C_1)$
m-word	`BBcbCBBabABbAaBBbCcBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3), \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

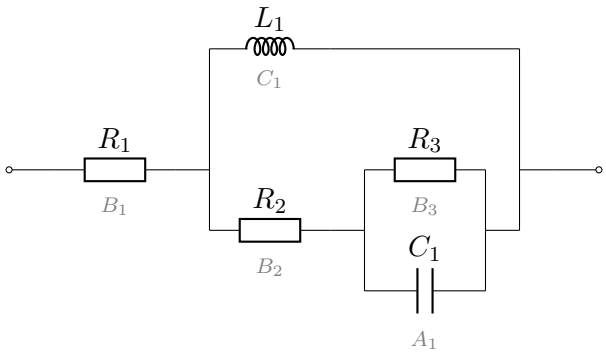
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1) \,\mathsf{p}\, C_1$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, ((B_3 \,\mathsf{s}\, A_1) \,\mathsf{p}\, C_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1 \,\mathsf{p}\, C_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
C_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, C_1 short$\}$ $\{B_3$ open, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_2$ short, C_1 short$\}$ $\{B_3$ open, A_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_2 open, B_3 short$\}$ $\{B_2$ short, C_1 short$\}$ $\{B_3$ open, A_1 short, A_1 open, C_1 open$\}$

Network 69



Network

realisation	`R s (L p (R s (R p C)))`
m-expression	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)))$
m-word	`BBcbCCbcBBBBBabAABBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0,2,2,0)$, $\quad Z(s) = s^0\, P/Q$, $\quad \deg P = 2$, $\quad \deg Q = 2$, $\quad \lambda = 4$
corners	$\delta_0 = B_1$, $\quad \delta_\infty = B_1 \,\mathsf{s}\, B_2$, $\quad \delta_\mathrm{K} = B_1$, $\quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

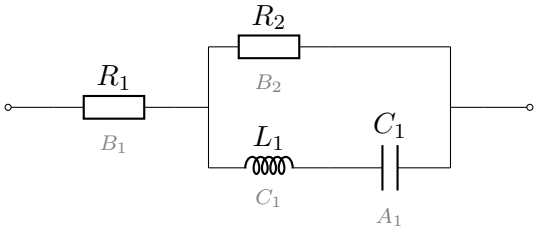
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$	`CAB`	$(1,2,1,0)$	DC, sweep	
B_1	open	P	`P`	–	port	
C_1	short	B_1	`B`	$(0,0,0,0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)$	`BAB`	$(0,1,1,0)$	sweep	
B_2	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_1)$	`BCAB`	$(0,2,2,0)$	none	✓
B_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0,0,1,1)$	HF, sweep	
B_3	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_2)$	`BCB`	$(0,1,1,0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1))$	`BCAB`	$(0,2,2,0)$	none	✓
A_1	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, B_2)$	`BCB`	$(0,1,1,0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3))$	`BCB`	$(0,1,1,0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_3 short, A_1 short, A_1 open}
absolute (identical residual networks)	{B_3 short, A_1 short}
signature (same quadruple)	{*nominal*, B_2 short, B_3 open} {C_1 open, B_3 short, A_1 short, A_1 open}

Network 71



Network

realisation	`R s (R p (L s C))`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, A_1))$
m-word	`BBBBBabACcAaCBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_2, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

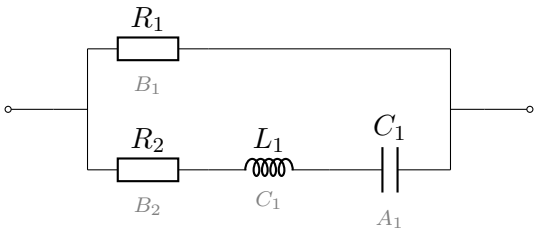
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, A_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1 \,\mathsf{s}\, A_1$	`AC`	$(-1, 0, 2, 1)$	DC, HF, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, C_1 open, A_1 open}
absolute (identical residual networks)	{C_1 open, A_1 open}
signature (same quadruple)	{*nominal*, B_1 short} {B_2 short, C_1 open, A_1 open} {C_1 short, A_1 short}

Network 72



Network

realisation	`R p (R s L s C)`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1 \,\mathsf{s}\, A_1)$
m-word	`BBabABBcbCCcAaCB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

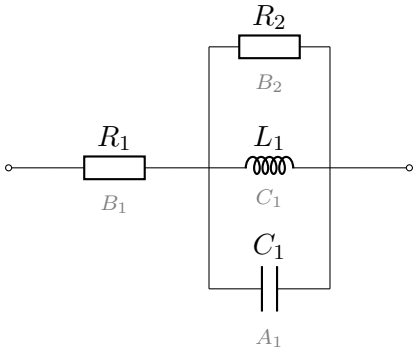
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, C_1 \,\mathsf{s}\, A_1$	`AC`	$(-1, 0, 2, 1)$	DC, HF, sweep	
B_2	short	$B_1 \,\mathsf{p}\, (C_1 \,\mathsf{s}\, A_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, C_1 open, A_1 open}
absolute (identical residual networks)	{B_2 open, C_1 open, A_1 open}
signature (same quadruple)	{*nominal*, B_2 short} {B_2 open, C_1 open, A_1 open} {C_1 short, A_1 short}

Network 73



Network

realisation	`R s (R p L p C)`
m-expression	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_1 \,\mathtt{p}\, A_1)$
m-word	`BBcbCCBbCcBabAAB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathtt{s}\, B_2$

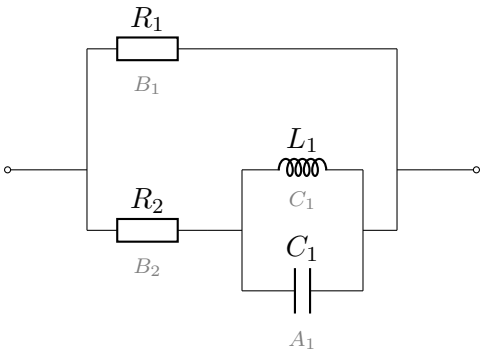
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathtt{p}\, C_1 \,\mathtt{p}\, A_1$	`CA`	$(1, 2, 0, -1)$	DC, HF, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathtt{s}\, (C_1 \,\mathtt{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
C_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathtt{s}\, (B_2 \,\mathtt{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, C_1 short, A_1 short}
absolute (identical residual networks)	{B_2 short, C_1 short, A_1 short}
signature (same quadruple)	{*nominal*, B_2 open} {B_2 short, C_1 short, A_1 short} {C_1 open, A_1 open}

Network 74



Network

realisation	`R p (R s (L p C))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, A_1))$
m-word	`BBBBBcbCCacAABB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

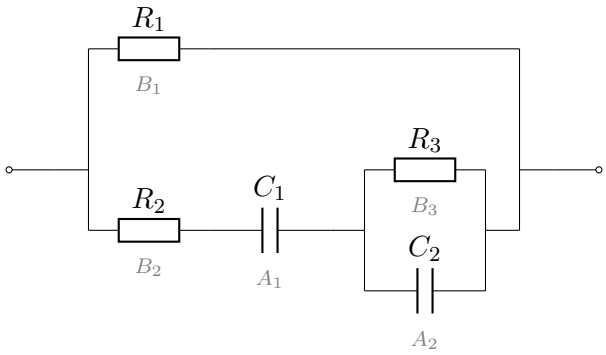
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, A_1$	`CA`	$(1, 2, 0, -1)$	DC, HF, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, C_1 short, A_1 short}
absolute (identical residual networks)	{C_1 short, A_1 short}
signature (same quadruple)	{*nominal*, B_1 open} {B_2 open, C_1 short, A_1 short} {C_1 open, A_1 open}

Network 75



Network

realisation	`R p ( R s C s (R p C))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2))$
m-word	`BBabAABbAaBBBBabAABB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

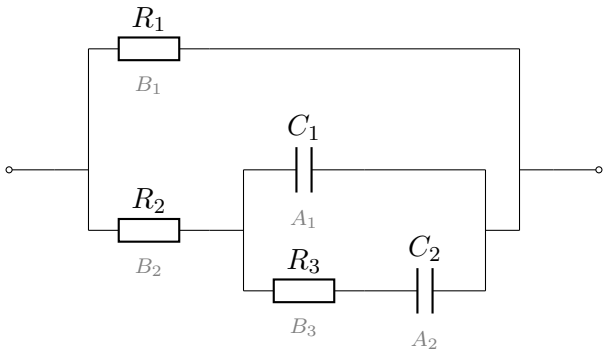
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2))$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2))$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, A_1 open$\}$ $\{B_3$ short, B_3 open, A_2 short, A_2 open$\}$
absolute (identical residual networks)	$\{B_2$ open, A_1 open$\}$ $\{B_3$ short, A_2 short$\}$
signature (same quadruple)	$\{B_2$ open, A_1 open$\}$ $\{A_1$ short, B_3 short, B_3 open, A_2 short, A_2 open$\}$

Network 76



Network

realisation	`R p (R s (C p (R s C)))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2)))$
m-word	`BBabABbAAAAaBbAaBABB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

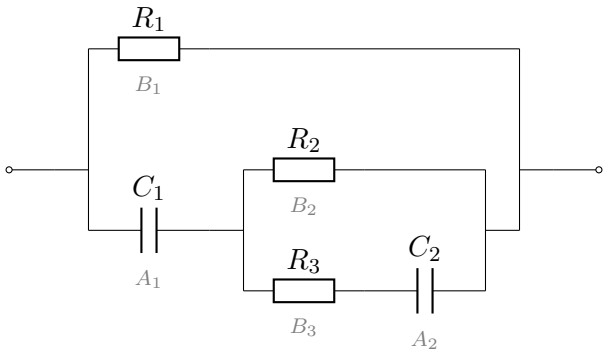
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2))$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2)$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, A_2))$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, A_1 short$\}$ $\{A_1$ open, B_3 short, B_3 open, A_2 open$\}$
absolute (identical residual networks)	$\{B_3$ open, A_2 open$\}$
signature (same quadruple)	$\{B_2$ open, A_1 short$\}$ $\{A_1$ open, B_3 short, B_3 open, A_2 short, A_2 open$\}$

Network 77



Network

realisation	`R p (C s (R p (R s C)))`
m-expression	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2)))$
m-word	`BBabAAbaBBabABbAaBBBB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1$

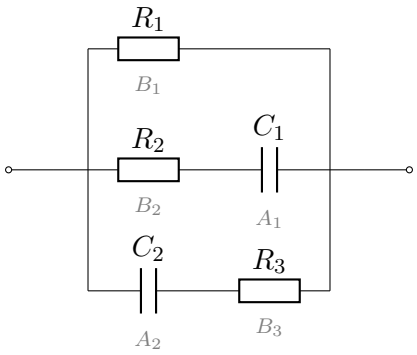
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2))$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2))$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_3	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{A_1 short, B_2 open, B_3 open, A_2 short, A_2 open}
absolute (identical residual networks)	{B_3 open, A_2 open}
signature (same quadruple)	{A_1 short, B_2 open, B_3 open, A_2 short, A_2 open}

Network 78



Network

realisation	`R p (R s C) p (C s R)`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, (A_2 \,\mathsf{s}\, B_3)$
m-word	`BBBabABbAaBBabAAbaBBB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1$

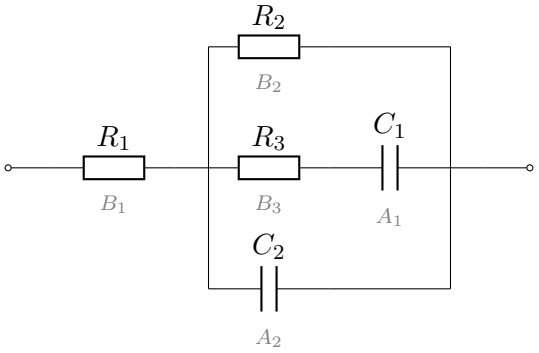
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, (A_2 \,\mathsf{s}\, B_3)$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1 \,\mathsf{p}\, (A_2 \,\mathsf{s}\, B_3)$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (A_2 \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (A_2 \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (A_2 \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, B_3$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, B_3 short$\}$ $\{B_2$ open, A_1 short, A_1 open, A_2 short, A_2 open, B_3 open$\}$
absolute (identical residual networks)	$\{B_2$ open, A_1 open$\}$ $\{A_2$ open, B_3 open$\}$
signature (same quadruple)	$\{B_2$ short, B_3 short$\}$ $\{B_2$ open, A_1 short, A_1 open, A_2 short, A_2 open, B_3 open$\}$

Network 79



Network

realisation	`R s (R p (R s C) p C)`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2)$
m-word	`BBBBBBabABbAaBBabAAB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

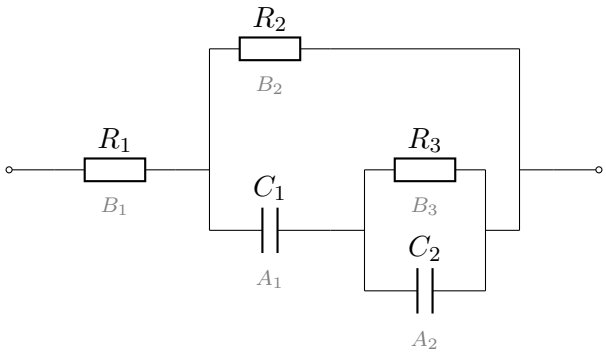
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, ((B_3 \,\mathsf{s}\, A_1) \,\mathsf{p}\, A_2)$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, A_2 short$\}$ $\{B_3$ short, B_3 open, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_2$ short, A_2 short$\}$ $\{B_3$ open, A_1 open$\}$
signature (same quadruple)	$\{B_2$ short, A_2 short$\}$ $\{B_3$ short, B_3 open, A_1 short, A_1 open, A_2 open$\}$

Network 80



Network

realisation	`R s (R p (C s (R p C)))`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)))$
m-word	`BBBBBabAAbaBBabAAAAB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

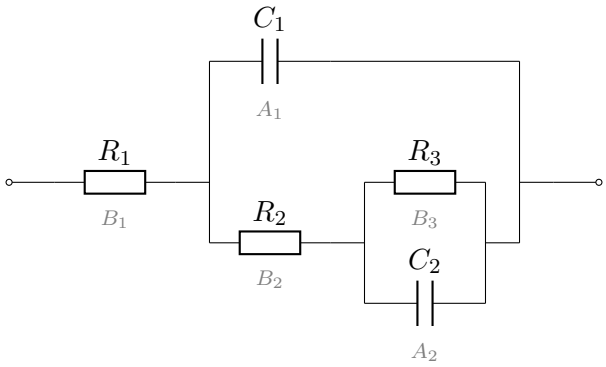
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2))$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, A_2))$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, A_1 open$\}$ $\{A_1$ short, B_3 short, B_3 open, A_2 short$\}$
absolute (identical residual networks)	$\{B_3$ short, A_2 short$\}$
signature (same quadruple)	$\{B_2$ short, A_1 open$\}$ $\{A_1$ short, B_3 short, B_3 open, A_2 short, A_2 open$\}$

Network 81



Network

realisation	`R s (C p (R s (R p C)))`
m-expression	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)))$
m-word	`BBBBAaBBBBBbabAABAB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

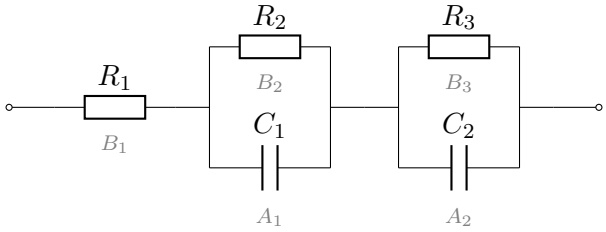
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2))$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_3	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_2))$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_2	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{A_1 open, B_2 short, B_3 short, A_2 short, A_2 open}
absolute (identical residual networks)	{B_3 short, A_2 short}
signature (same quadruple)	{A_1 open, B_2 short, B_3 short, A_2 short, A_2 open}

Network 82



Network

realisation	`R s (R p C) s (R p C)`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$
m-word	`BBBBBBabAABBBBabAAB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$, $\delta_\infty = B_1$, $\delta_\mathrm{K} = B_1$, $\delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

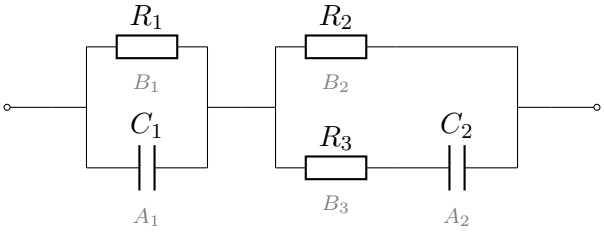
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$(B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, A_2$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_2	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_3$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, A_1 short, A_1 open, B_3 short, A_2 short, A_2 open} {B_2 open, B_3 open}
absolute (identical residual networks)	{B_2 short, A_1 short} {B_3 short, A_2 short}
signature (same quadruple)	{B_2 short, A_1 short, A_1 open, B_3 short, A_2 short, A_2 open} {B_2 open, B_3 open}

Network 83



Network

realisation	`(R p C) s (R p (R s C))`
m-expression	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2))$
m-word	`BBBabAAbaBBabABbAaBBB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2$, $\delta_\infty = B_2 \,\mathsf{p}\, B_3$, $\delta_\mathrm{K} = B_2 \,\mathsf{p}\, B_3$, $\delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

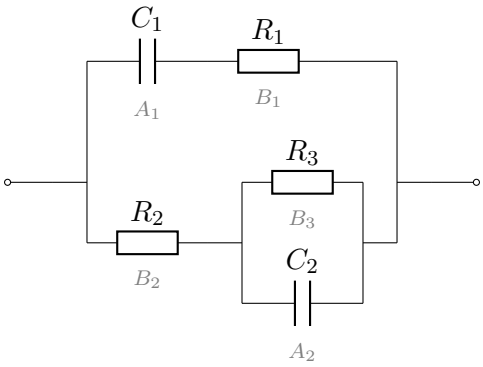
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2))$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_2))$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_2	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_2$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
B_3	short	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_2)$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_3	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	short	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_1 short$\}$ $\{B_3$ open, A_2 short, A_2 open$\}$
absolute (identical residual networks)	$\{B_1$ short, A_1 short$\}$ $\{B_3$ open, A_2 open$\}$
signature (same quadruple)	$\{B_1$ short, A_1 short, A_1 open, B_3 open, A_2 short, A_2 open$\}$ $\{B_1$ open, B_2 open$\}$

Network 84



Network

realisation	`(C s R) p (R s (R p C))`
m-expression	$(A_1 \mathsf{s} B_1) \mathsf{p} (B_2 \mathsf{s} (B_3 \mathsf{p} A_2))$
m-word	`BAAbaBBBBBBBBabAABB` $\rightarrow$ `BABAB`
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_2 \mathsf{s} B_3, \quad \delta_\infty = B_1 \mathsf{p} B_2, \quad \delta_\mathrm{K} = B_1 \mathsf{p} B_2, \quad \delta_\mathrm{P} = B_2 \mathsf{s} B_3$

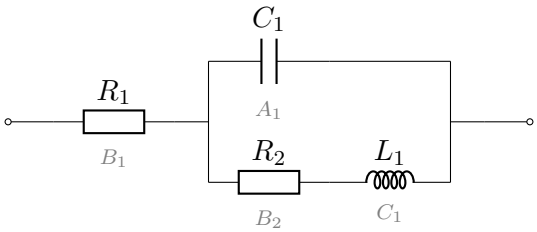
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	$B_1 \mathsf{p} (B_2 \mathsf{s} (B_3 \mathsf{p} A_2))$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_2 \mathsf{s} (B_3 \mathsf{p} A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_1	short	$A_1 \mathsf{p} (B_2 \mathsf{s} (B_3 \mathsf{p} A_2))$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	$B_2 \mathsf{s} (B_3 \mathsf{p} A_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$(A_1 \mathsf{s} B_1) \mathsf{p} B_3 \mathsf{p} A_2$	`BABA`	$(0, 2, 1, -1)$	HF, sweep	
B_2	open	$A_1 \mathsf{s} B_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_3	short	$(A_1 \mathsf{s} B_1) \mathsf{p} B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$(A_1 \mathsf{s} B_1) \mathsf{p} (B_2 \mathsf{s} A_2)$	`ABAB`	$(-1, 1, 2, 0)$	DC, sweep	
A_2	short	$(A_1 \mathsf{s} B_1) \mathsf{p} B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_2	open	$(A_1 \mathsf{s} B_1) \mathsf{p} (B_2 \mathsf{s} B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{A_1$ open, B_1 open$\}$ $\{B_3$ short, A_2 short, A_2 open$\}$
absolute (identical residual networks)	$\{A_1$ open, B_1 open$\}$ $\{B_3$ short, A_2 short$\}$
signature (same quadruple)	$\{A_1$ short, A_1 open, B_1 open, B_3 short, A_2 short, A_2 open$\}$ $\{B_1$ short, B_2 short$\}$

Network 87



Network

realisation	`R s (C p (R s L))`
m-expression	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1))$
m-word	`BBBBAaBBbcbCCAB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = \mathrm{P}$

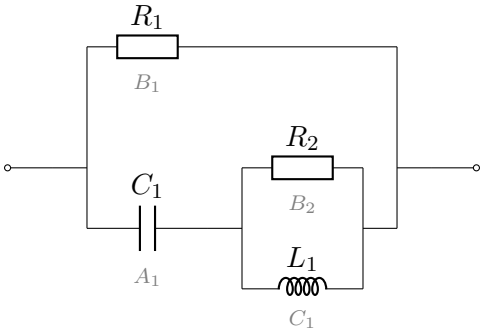
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
B_2	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, C_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, C_1 open$\}$
absolute (identical residual networks)	$\{B_2$ open, C_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_2 short$\}$ $\{B_2$ open, C_1 open$\}$

Network 88



Network

realisation	`R p (C s (R p L))`
m-expression	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1))$
m-word	`BBabAAcaCBbCcBBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1$, $\delta_\infty = B_1 \,\mathsf{p}\, B_2$, $\delta_\mathrm{K} = \mathrm{K}$, $\delta_\mathrm{P} = B_1$

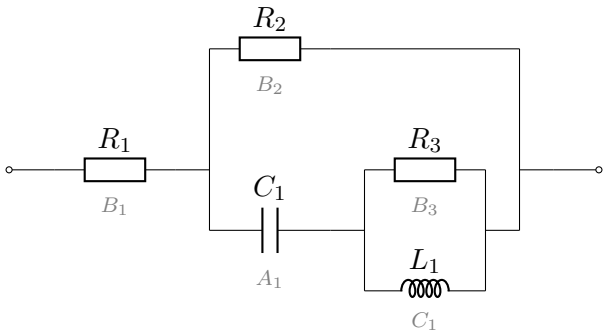
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
C_1	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
C_1	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 short, C_1 short}
absolute (identical residual networks)	{B_2 short, C_1 short}
signature (same quadruple)	{*nominal*, B_2 open} {B_2 short, C_1 short}

Network 89



Network

realisation	`R s (R p (C s (R p L)))`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)))$
m-word	`BBBBBabAAcaCBbCcBBBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3), \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

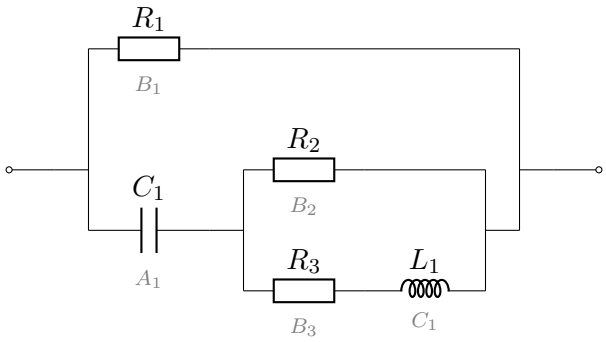
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, C_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, A_1 open$\}$ $\{B_3$ short, C_1 short$\}$
absolute (identical residual networks)	$\{B_3$ short, C_1 short$\}$
signature (same quadruple)	$\{$*nominal*, B_1 short, B_3 open$\}$ $\{B_2$ short, A_1 open$\}$ $\{A_1$ short, B_3 short, C_1 short, C_1 open$\}$

Network 90



Network

realisation	`R p (C s (R p (R s L)))`
m-expression	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1)))$
m-word	`BBabAAbaBBBBBcbCCBBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1$, $\delta_\infty = B_1 \,\mathsf{p}\, B_2$, $\delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3$, $\delta_\mathrm{P} = B_1$

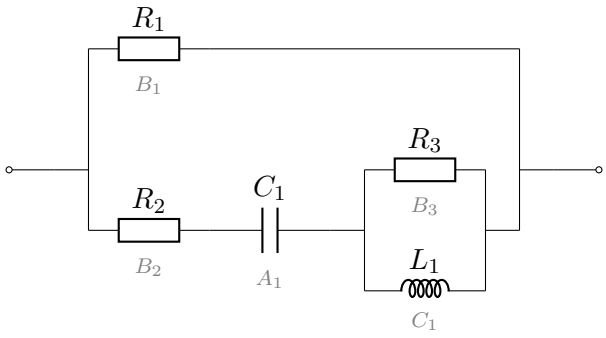
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1))$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_3 \,\mathsf{s}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_3	short	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_3 open, C_1 short, C_1 open}
absolute (identical residual networks)	{B_3 open, C_1 open}
signature (same quadruple)	{*nominal*, B_2 open, B_3 short}
	{A_1 short, B_3 open, C_1 short, C_1 open}

Network 91



Network

realisation	`R p (R s C s (R p L))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$
m-word	`BBabAABbAaBcbCBbCcBBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1$, $\delta_\infty = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$, $\delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2$, $\delta_\mathrm{P} = B_1$

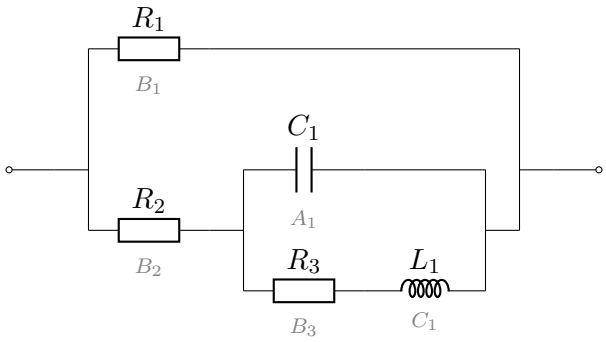
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
B_2	short	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1 \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, A_1 open} {B_3 short, C_1 short, C_1 open}
absolute (identical residual networks)	{B_2 open, A_1 open} {B_3 short, C_1 short}
signature (same quadruple)	{*nominal*, B_2 short, B_3 open} {B_2 open, A_1 open} {A_1 short, B_3 short, C_1 short, C_1 open}

Network 92



Network

realisation	`R p (R s (C p (R s L)))`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1)))$
m-word	`BBBBBBBAaBBbcbCCABB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3), \quad \delta_\infty = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_1$

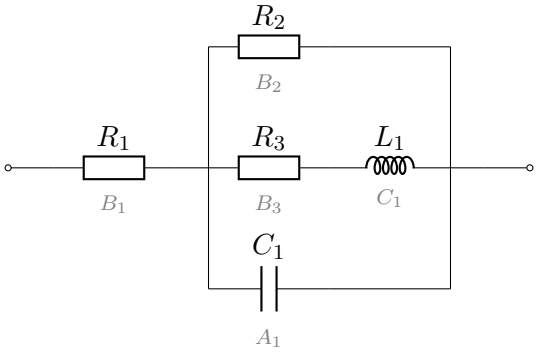
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1))$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, A_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1)$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_2	open	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, B_2$	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, C_1))$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, A_1 short$\}$ $\{B_3$ open, C_1 open$\}$
absolute (identical residual networks)	$\{B_3$ open, C_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_1 open, B_3 short$\}$ $\{B_2$ open, A_1 short$\}$ $\{A_1$ open, B_3 open, C_1 short, C_1 open$\}$

Network 93



Network

realisation	`R s (R p (R s L) p C)`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1)$
m-word	`BBBBBBBBBcbCCBabAAB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3), \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

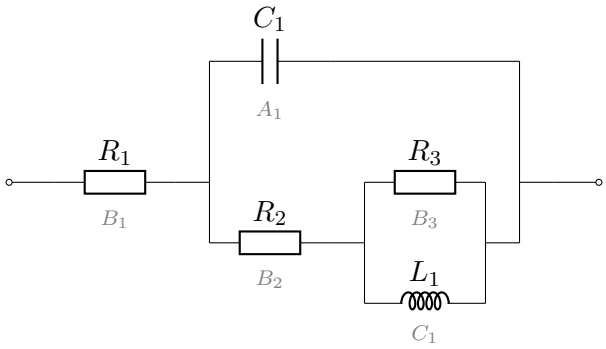
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
B_2	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, ((B_3 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1 \,\mathsf{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1))$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, A_1 short$\}$ $\{B_3$ open, C_1 short, C_1 open$\}$
absolute (identical residual networks)	$\{B_2$ short, A_1 short$\}$ $\{B_3$ open, C_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_2 open, B_3 short$\}$ $\{B_2$ short, A_1 short$\}$ $\{B_3$ open, C_1 short, C_1 open, A_1 open$\}$

Network 94



Network

realisation	`R s (C p (R s (R p L)))`
m-expression	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)))$
m-word	`BBBBAaBBbcbCBbCcBBAB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

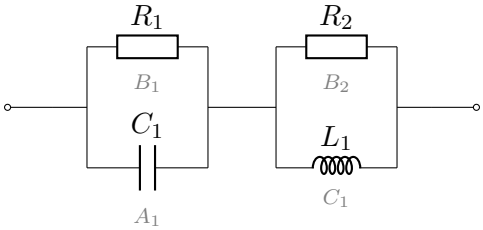
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	P	`P`	–	port	
A_1	short	B_1	`B`	$(0, 0, 0, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_3	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1))$	`BCAB`	$(0, 2, 2, 0)$	none	✓
C_1	short	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, B_2)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3))$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_3 short, C_1 short, C_1 open}
absolute (identical residual networks)	{B_3 short, C_1 short}
signature (same quadruple)	{*nominal*, B_2 short, B_3 open} {A_1 open, B_3 short, C_1 short, C_1 open}

Network 96



Network

realisation	`(R p C) s (R p L)`
m-expression	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$
m-word	`BBBabAAcaCBbCcBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1, \quad \delta_\infty = B_2, \quad \delta_\mathrm{K} = \mathrm{K}, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

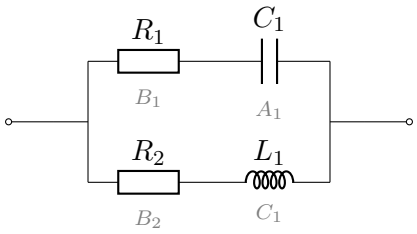
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_2 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_2	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, C_1$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
C_1	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_1 short$\}$ $\{B_2$ short, C_1 short$\}$
absolute (identical residual networks)	$\{B_1$ short, A_1 short$\}$ $\{B_2$ short, C_1 short$\}$
signature (same quadruple)	$\{B_1$ short, A_1 short$\}$ $\{A_1$ open, C_1 open$\}$ $\{B_2$ short, C_1 short$\}$

Network 97



Network

realisation	`(R s C) p (R s L)`
m-expression	$(B_1 \,\mathsf{s}\, A_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$
m-word	`BABbAaBBBBcbCCB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_2, \quad \delta_\infty = B_1, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = \mathrm{P}$

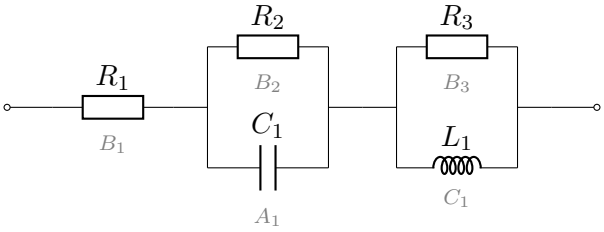
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	$B_2 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_2 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
B_2	short	$(B_1 \,\mathsf{s}\, A_1) \,\mathsf{p}\, C_1$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
C_1	short	$(B_1 \,\mathsf{s}\, A_1) \,\mathsf{p}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, A_1 open$\}$ $\{B_2$ open, C_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, A_1 open$\}$ $\{B_2$ open, C_1 open$\}$
signature (same quadruple)	$\{B_1$ open, A_1 open$\}$ $\{A_1$ short, C_1 short$\}$ $\{B_2$ open, C_1 open$\}$

Network 98



Network

realisation	`R s (R p C) s (R p L)`
m-expression	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$
m-word	`BBBBBBabAABcbCBbCcBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2, \quad \delta_\infty = B_1 \,\mathsf{s}\, B_3, \quad \delta_\mathrm{K} = B_1, \quad \delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, B_3$

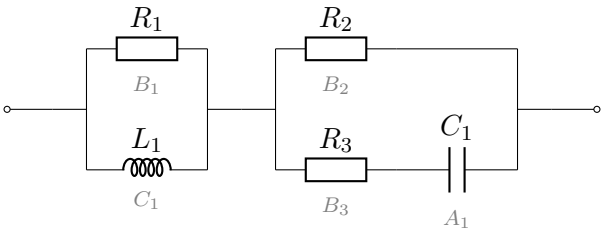
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$(B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	open	$B_1 \,\mathsf{s}\, A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, C_1$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_3$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, A_1 short, A_1 open$\}$ $\{B_3$ short, C_1 short, C_1 open$\}$
absolute (identical residual networks)	$\{B_2$ short, A_1 short$\}$ $\{B_3$ short, C_1 short$\}$
signature (same quadruple)	$\{$*nominal*, B_1 short$\}$ $\{B_2$ short, A_1 short, A_1 open, B_3 short, C_1 short, C_1 open$\}$

Network 99



Network

realisation	`(R p L) s (R p (R s C))`
m-expression	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$
m-word	`BCBbCcBBBBabABbAaBBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_2$, $\delta_\infty = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$, $\delta_\mathrm{K} = B_2 \,\mathsf{p}\, B_3$, $\delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

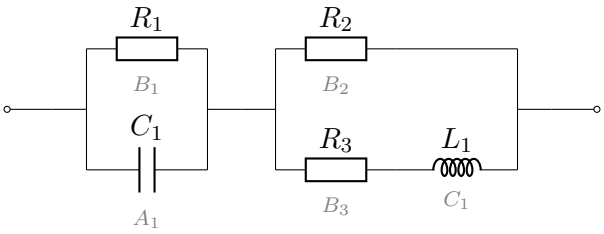
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_1	open	$C_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
C_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_2	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_1$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
B_3	short	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$(B_1 \,\mathsf{p}\, C_1) \,\mathsf{s}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{B_1 short, C_1 short} {B_3 open, A_1 short, A_1 open}
absolute (identical residual networks)	{B_1 short, C_1 short} {B_3 open, A_1 open}
signature (same quadruple)	{*nominal*, B_3 short} {B_1 short, C_1 short, C_1 open, B_3 open, A_1 short, A_1 open}

Network 100



Network

realisation	`(R p C) s (R p (R s L))`
m-expression	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1))$
m-word	`BBBabAAbaBBBBBcbCCBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$, $\delta_\infty = B_2$, $\delta_\mathrm{K} = B_2 \,\mathsf{p}\, B_3$, $\delta_\mathrm{P} = B_1 \,\mathsf{s}\, B_2$

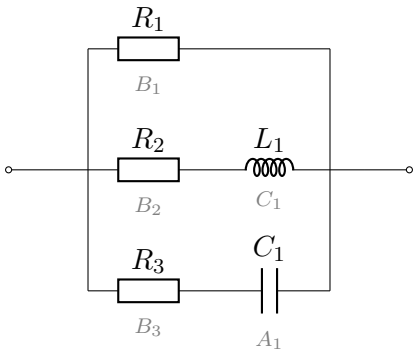
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_1	open	$A_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1))$	`ACB`	$(-1, 1, 2, 0)$	DC, sweep	
A_1	short	$B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, C_1))$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$B_1 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_2	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_3 \,\mathsf{s}\, C_1$	`BAC`	$(0, 1, 2, 1)$	HF, sweep	
B_3	short	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$(B_1 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ short, A_1 short$\}$ $\{B_3$ open, C_1 short, C_1 open$\}$
absolute (identical residual networks)	$\{B_1$ short, A_1 short$\}$ $\{B_3$ open, C_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_3 short$\}$ $\{B_1$ short, A_1 short, A_1 open, B_3 open, C_1 short, C_1 open$\}$

Network 101



Network

realisation	`R p (R s L) p (R s C)`
m-expression	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$
m-word	`BBBBBBcbCCBabABbAaBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2, \quad \delta_\infty = B_1 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, B_3, \quad \delta_\mathrm{P} = B_1$

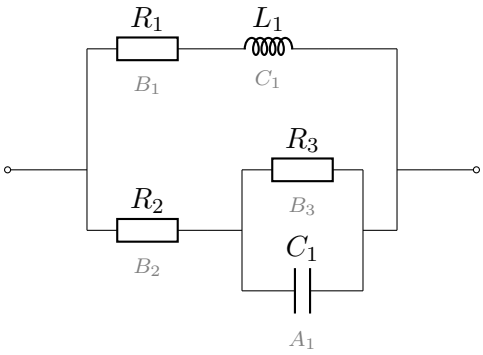
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_3	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_3$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ open, C_1 short, C_1 open$\}$ $\{B_3$ open, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_2$ open, C_1 open$\}$ $\{B_3$ open, A_1 open$\}$
signature (same quadruple)	$\{$*nominal*, B_1 open$\}$ $\{B_2$ open, C_1 short, C_1 open, B_3 open, A_1 short, A_1 open$\}$

Network 102



Network

realisation	`(R s L) p ( R s (R p C))`
m-expression	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$
m-word	`BBBcbCCbcBBBBBabAABB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0), \quad Z(s) = s^0\, P/Q, \quad \deg P = 2, \quad \deg Q = 2, \quad \lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3), \quad \delta_\infty = B_2, \quad \delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2, \quad \delta_\mathrm{P} = B_2 \,\mathsf{s}\, B_3$

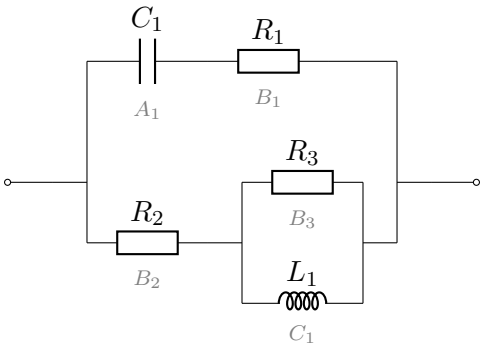
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$C_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_1	open	$B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1))$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, A_1)$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_3 \,\mathsf{p}\, A_1$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_2	open	$B_1 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
B_3	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
A_1	short	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_2$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$(B_1 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_1$ open, C_1 open$\}$ $\{B_3$ short, A_1 short, A_1 open$\}$
absolute (identical residual networks)	$\{B_1$ open, C_1 open$\}$ $\{B_3$ short, A_1 short$\}$
signature (same quadruple)	$\{$*nominal*, B_3 open$\}$ $\{B_1$ open, C_1 short, C_1 open, B_3 short, A_1 short, A_1 open$\}$

Network 103



Network

realisation	`(C s R) p (R s (R p L))`
m-expression	$(A_1 \,\mathsf{s}\, B_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$
m-word	`BAAbaBBBBBcbCBbCcBBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_2$, $\delta_\infty = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$, $\delta_\mathrm{K} = B_1 \,\mathsf{p}\, B_2$, $\delta_\mathrm{P} = B_2 \,\mathsf{s}\, B_3$

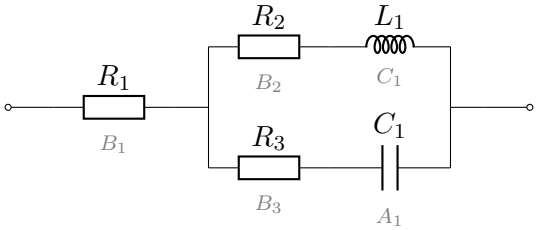
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
A_1	short	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_1	short	$A_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BCA`	$(0, 2, 1, -1)$	HF, sweep	
B_1	open	$B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_2	short	$(A_1 \,\mathsf{s}\, B_1) \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_1$	`CAB`	$(1, 2, 1, 0)$	DC, sweep	
B_2	open	$A_1 \,\mathsf{s}\, B_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_3	short	$(A_1 \,\mathsf{s}\, B_1) \,\mathsf{p}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
B_3	open	$(A_1 \,\mathsf{s}\, B_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
C_1	short	$(A_1 \,\mathsf{s}\, B_1) \,\mathsf{p}\, B_2$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$(A_1 \,\mathsf{s}\, B_1) \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	{A_1 open, B_1 open} {B_3 short, C_1 short, C_1 open}
absolute (identical residual networks)	{A_1 open, B_1 open} {B_3 short, C_1 short}
signature (same quadruple)	{*nominal*, B_3 open} {A_1 short, A_1 open, B_1 open, B_3 short, C_1 short, C_1 open}

Network 104



Network

realisation	`R s ((R s L) p (R s C))`
m-expression	$B_1 \,\mathsf{s}\, ((B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$
m-word	`BBBBBBcbCCacABbAaBBB` $\rightarrow$ `BCAB` (= `BACB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{s}\, B_2$, $\delta_\infty = B_1 \,\mathsf{s}\, B_3$, $\delta_\mathrm{K} = B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, B_3)$, $\delta_\mathrm{P} = \mathrm{P}$

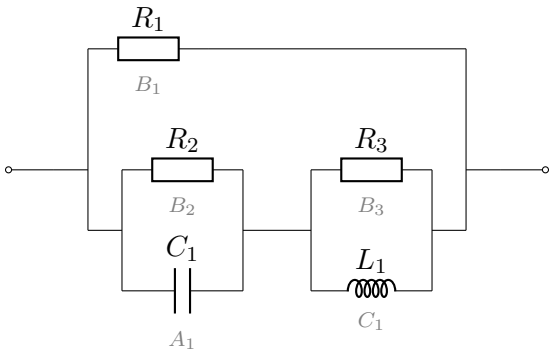
Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	$(B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_1	open	P	`P`	–	port	
B_2	short	$B_1 \,\mathsf{s}\, (C_1 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_2	open	$B_1 \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
C_1	short	$B_1 \,\mathsf{s}\, (B_2 \,\mathsf{p}\, (B_3 \,\mathsf{s}\, A_1))$	`BAB`	$(0, 1, 1, 0)$	sweep	
C_1	open	$B_1 \,\mathsf{s}\, B_3 \,\mathsf{s}\, A_1$	`AB`	$(-1, 0, 1, 0)$	DC, sweep	
B_3	short	$B_1 \,\mathsf{s}\, ((B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, A_1)$	`BCAB`	$(0, 2, 2, 0)$	none	✓
B_3	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	
A_1	short	$B_1 \,\mathsf{s}\, ((B_2 \,\mathsf{s}\, C_1) \,\mathsf{p}\, B_3)$	`BCB`	$(0, 1, 1, 0)$	sweep	
A_1	open	$B_1 \,\mathsf{s}\, B_2 \,\mathsf{s}\, C_1$	`BC`	$(0, 0, 1, 1)$	HF, sweep	

Ambiguity groups

structural (same topology after merging)	{B_2 open, C_1 open} {B_3 open, A_1 open}
absolute (identical residual networks)	{B_2 open, C_1 open} {B_3 open, A_1 open}
signature (same quadruple)	{*nominal*, B_1 short, B_2 short, B_3 short} {B_2 open, C_1 open} {C_1 short, A_1 short} {B_3 open, A_1 open}

Network 106



Network

realisation	`R p ((R p C) s (R p L))`
m-expression	$B_1 \,\mathsf{p}\, ((B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$
m-word	`BBBBBBabAAcaCBbCcBBB` $\rightarrow$ `BACB` (= `BCAB` by the reversal rule)
signature	$\Phi = (0, 2, 2, 0)$, $Z(s) = s^0\, P/Q$, $\deg P = 2$, $\deg Q = 2$, $\lambda = 4$
corners	$\delta_0 = B_1 \,\mathsf{p}\, B_2$, $\delta_\infty = B_1 \,\mathsf{p}\, B_3$, $\delta_\mathrm{K} = \mathrm{K}$, $\delta_\mathrm{P} = B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, B_3)$

Single hard faults

element	fault	faulty network	m-word	Φ	detected by	str. undet.
B_1	short	K	`K`	–	port	
B_1	open	$(B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
B_2	short	$B_1 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
B_2	open	$B_1 \,\mathsf{p}\, (A_1 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BACB`	$(0, 2, 2, 0)$	none	✓
A_1	short	$B_1 \,\mathsf{p}\, B_3 \,\mathsf{p}\, C_1$	`CB`	$(1, 1, 0, 0)$	DC, sweep	
A_1	open	$B_1 \,\mathsf{p}\, (B_2 \,\mathsf{s}\, (B_3 \,\mathsf{p}\, C_1))$	`BCB`	$(0, 1, 1, 0)$	sweep	
B_3	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
B_3	open	$B_1 \,\mathsf{p}\, ((B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, C_1)$	`BACB`	$(0, 2, 2, 0)$	none	✓
C_1	short	$B_1 \,\mathsf{p}\, B_2 \,\mathsf{p}\, A_1$	`BA`	$(0, 1, 0, -1)$	HF, sweep	
C_1	open	$B_1 \,\mathsf{p}\, ((B_2 \,\mathsf{p}\, A_1) \,\mathsf{s}\, B_3)$	`BAB`	$(0, 1, 1, 0)$	sweep	

Ambiguity groups

structural (same topology after merging)	$\{B_2$ short, A_1 short$\}$ $\{B_3$ short, C_1 short$\}$
absolute (identical residual networks)	$\{B_2$ short, A_1 short$\}$ $\{B_3$ short, C_1 short$\}$
signature (same quadruple)	$\{$*nominal*, B_1 open, B_2 open, B_3 open$\}$ $\{B_2$ short, A_1 short$\}$ $\{A_1$ open, C_1 open$\}$ $\{B_3$ short, C_1 short$\}$